SAT Math 1.1 — Linear Equations in One Variable

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Multi-step equations

undo the operations in reverse order | whatever you do to one side, do to the other

Work from the outside in: deal with addition and subtraction before multiplication and division. Substituting your answer back is a ten-second check that catches almost every slip.



then check: 5(8 − 2) = 30 ✓

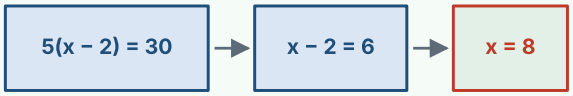
Each box undoes one operation, in the reverse of the order they were applied. The last box is the answer, and the check puts it straight back into the original.

Example 1. **Solve 5(x − 2) = 30.**

Solution.

Divide by 5: x − 2 = 6.

x = **8**. Check: 5(6) = 30.

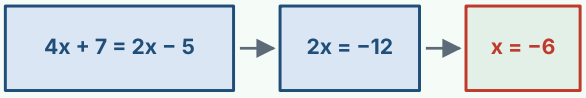


Example 2. **Solve 4x + 7 = 2x − 5.**

Solution.

Collect x on one side: 2x = −12.

x = **−6**.

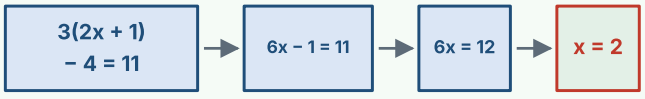


Example 3. **Solve 3(2x + 1) − 4 = 11.**

Solution.

6x + 3 − 4 = 11, so 6x = 12.

x = **2**.



2 Clearing fractions and decimals

multiply every term by the common denominator

Fractions make arithmetic slow and errors likely. Multiply the whole equation through by the lowest common denominator first, and the problem becomes an ordinary one.

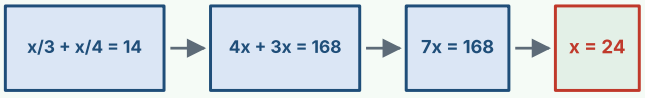


every term — including the one on the right

Twelve is the lowest number both 3 and 4 divide into, so one multiplication clears both fractions at once.

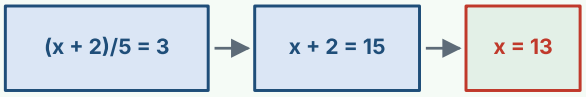
Example 1. **Solve x/3 + x/4 = 14.**

Solution.
Multiply everything by 12: 4x + 3x = 168.
7x = 168, so x = **24**.



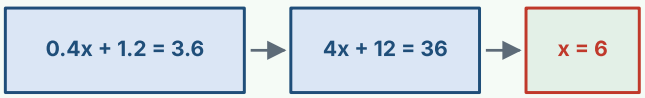
Example 2. **Solve (x + 2)/5 = 3.**

Solution.
Multiply by 5: x + 2 = 15.
x = **13**.



Example 3. **Solve 0.4x + 1.2 = 3.6.**

Solution.
Multiply by 10 to clear decimals: 4x + 12 = 36.
4x = 24, so x = **6**.



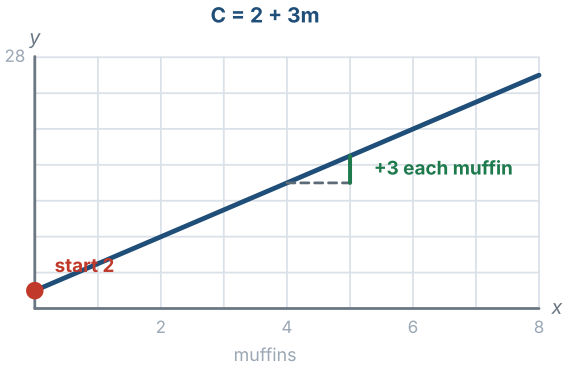
3 Translating a sentence

name the unknown | write one equation | answer the question that was asked

Most SAT algebra is a sentence in disguise. ‘A fixed charge plus so much each’ is always b + mx. Write the equation before touching any arithmetic.

the words map onto the symbols, phrase by phrase

a fixed charge plus so much each	b + mx
‘$2 for a box plus $3 per muffin’	2 + 3m
‘three more than twice a number’	2n + 3



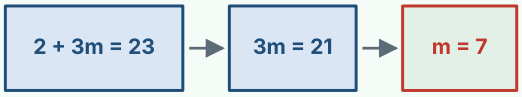
The fixed charge is where the line starts and the per-item charge is how fast it climbs. Paying $23 means asking which m puts you at that height.

Example 1. **A bakery charges $2 for a box plus $3 per muffin. A customer pays $23. How many muffins?**

Solution.

2 + 3m = 23, so 3m = 21.

m = **7 muffins**.



Example 2. **A fabric shop charges $4.50 per metre plus a $2 cutting fee. A customer pays $29. How many metres?**

Solution.

4.5x + 2 = 29, so 4.5x = 27.

x = **6 metres**.

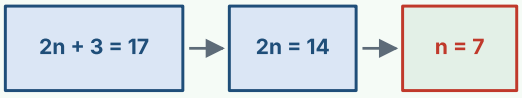


Example 3. **Three more than twice a number is 17. Find the number.**

Solution.

2n + 3 = 17, so 2n = 14.

n = **7**.



4 Literal equations

treat the other letters as numbers and isolate the one you want

Rearranging a formula uses exactly the same moves as solving for x. If the target letter appears twice, gather those terms together and factor it out.



the same three moves as solving for a number

when the letter appears twice, factor it out

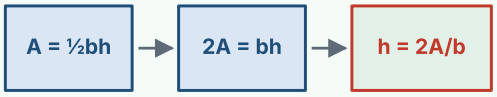
Top: isolate h exactly as you would isolate x. Bottom: with x on both sides you must collect first — dividing early is what loses the mark.

Example 1. **Solve A = ½bh for h.**

Solution.

Multiply by 2: 2A = bh.

h = **2A/b**.

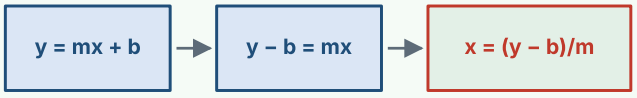


Example 2. **Solve y = mx + b for x.**

Solution.

y − b = mx.

x = **(y − b)/m**.

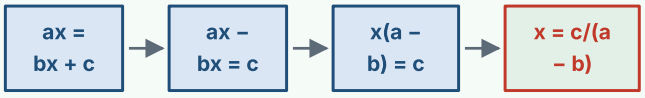


Example 3. **Solve ax = bx + c for x, where a ≠ b.**

Solution.

x appears twice, so gather and factor: ax − bx = c, giving x(a − b) = c.

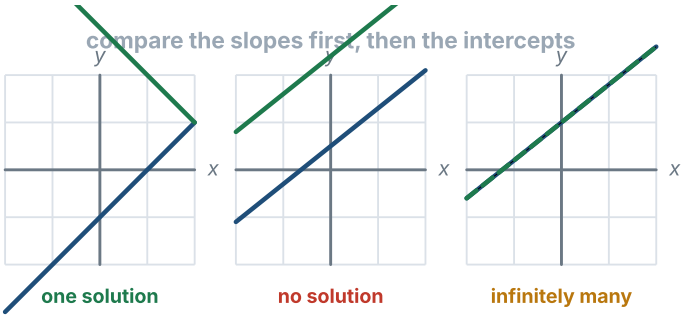
x = **c/(a − b)**.



5 No solution or infinitely many

variables cancel and the statement is false → no solution | true → infinitely many

Simplify both sides fully. If the x terms vanish, look at what is left: 5 = 7 is false, so nothing works; 5 = 5 is true, so everything works. For ax + b = cx + d this means a = c with b ≠ d, or a = c with b = d.



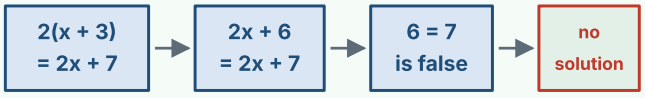
The same three cases drawn as lines: crossing once, running parallel with no crossing, or lying on top of each other so every point works.

Example 1. **Solve 2(x + 3) = 2x + 7.**

Solution.

2x + 6 = 2x + 7 gives 6 = 7, which is false.

No solution.

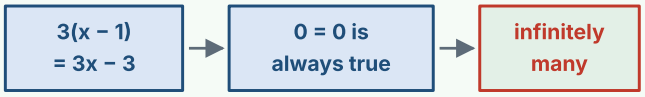


Example 2. **Solve 3(x − 1) = 3x − 3.**

Solution.

3x − 3 = 3x − 3 gives 0 = 0, which is always true.

Infinitely many solutions.



Example 3. **For what k does kx + 5 = 4x + 5 have infinitely many solutions?**

Solution.

The sides must match exactly.

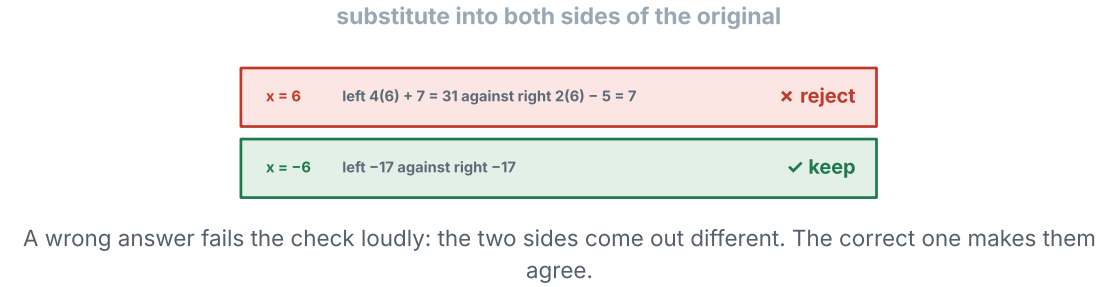
k = **4**.

kx + 5 = 4x + 5	the constants already match
so the x terms must too	k = 4

6 Checking and interpreting

substitute back | and make sure the answer is the quantity asked for

Two different habits, both worth marks. Substituting catches arithmetic errors. Re-reading the question catches the trap where you solve for x but the question wanted 2x, or the price rather than the quantity.

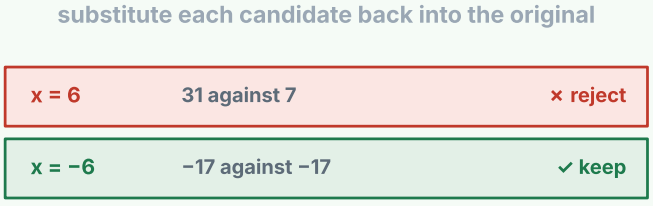


A wrong answer fails the check loudly: the two sides come out different. The correct one makes them agree.

Example 1. **A student solves 4x + 7 = 2x − 5 and gets x = 6. Check it.**

Solution.

Left: 31. Right: 7. They differ, so **x = 6 is wrong**; the correct answer is −6.

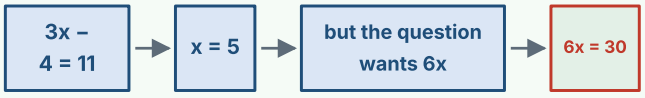


Example 2. **If 3x − 4 = 11, what is the value of 6x?**

Solution.

3x = 15, so x = 5.

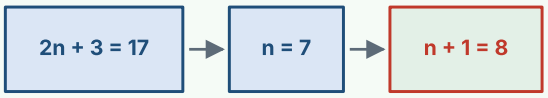
6x = **30**, not 5 — read what was asked.



Example 3. **If 2n + 3 = 17, what is n + 1?**

Solution.

n = 7, so n + 1 = **8**.



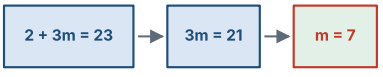
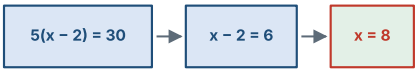
Part 1 — With Desmos (10 questions · every one is faster on the graph)

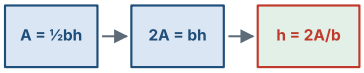
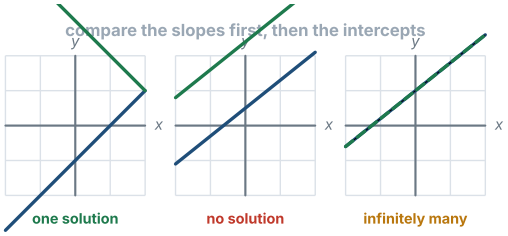
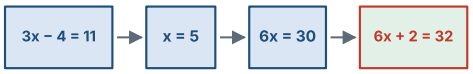
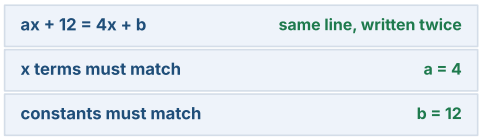
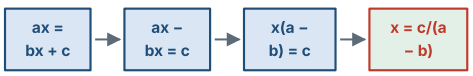
The SAT gives you Desmos on every maths question. Here it is **an equation solver**. Type the equation exactly as written and Desmos returns the value; the skill is turning the sentence into that equation.

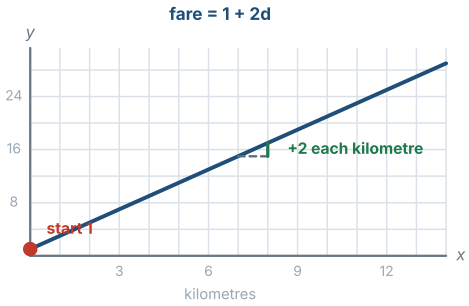
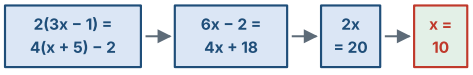
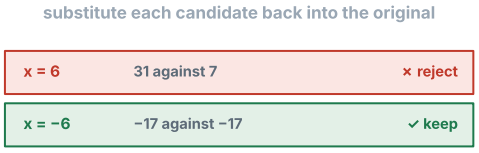
#	Question	Type into Desmos	Then	Answer
D1	If 5(x − 2) = 30, what is x?	`5(x-2)=30`	Type the equation; Desmos solves it.	8
D2	Solve 4x + 7 = 2x − 5.	`4x+7=2x-5`	Variables on both sides are no harder.	-6
D3	Solve x/3 + x/4 = 14.	`x/3+x/4=14`	Fractions are fine typed directly.	24
D4	A bakery charges $2 plus $3 per muffin and the total is $23. How many muffins?	`2+3m=23`	Write the model, then solve.	7
D5	A shop charges $4.50 per metre plus a $2 fee and the total is $29. How many metres?	`4.5x+2=29`	Fixed part plus rate times amount.	6
D6	Solve 0.4x + 1.2 = 3.6.	`0.4x+1.2=3.6`	Decimals need no clearing in Desmos.	6
D7	If 3x − 4 = 11, what is 6x?	`6*((11+4)/3)`	Solve, then answer what was asked.	30
D8	Solve 3(2x + 1) − 4 = 11.	`3(2x+1)-4=11`	Brackets are handled as typed.	2
D9	For what k does kx + 5 = 4x + 5 have infinitely many solutions?	`4`	The two sides must be identical.	4
D10	Solve (x + 2)/5 = 3.	`(x+2)/5=3`	One step after clearing the denominator.	13

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Several of these solve to a value that is not the answer — re-read the final sentence before writing anything down.

#	Level	Question	Answer	Solution
P1	Hard	A bakery charges $2 for a gift box plus $3 for each muffin placed in it. A customer pays $23 in total for one box of muffins. How many muffins are in the box? 	7	2 + 3m = 23, so 3m = 21. m = **7 muffins**. A common wrong answer comes from 23/3 rounded, ignoring the box charge.
P2	Hard	If 5(x − 2) = 30, what is the value of x? 	8	Divide both sides by 5 first: x − 2 = 6. x = **8**. A common wrong answer is x − 2, one step short of the answer.

#	Level	Question	Answer	Solution
P3	Hard	A fabric shop charges $4.50 per metre of linen plus a flat $2 cutting fee. A customer pays $29 in total. How many metres of linen did the customer buy? 	6	4.5x + 2 = 29 gives 4.5x = 27. x = **6 metres**.
P4	Hard	If x/3 + x/4 = 14, what is the value of x? 	24	Multiply every term by 12: 4x + 3x = 168, so 7x = 168. x = **24**.
P5	Hard	The equation A = ½bh gives the area of a triangle. Which expression gives h in terms of A and b? 	2A/b	Multiply both sides by 2, then divide by b. h = **2A/b**. A common wrong answer divides by 2 instead of multiplying.
P6	Hard	How many solutions does the equation 2(x + 3) = 2x + 7 have? 	Zero	Expanding gives 2x + 6 = 2x + 7, and the x terms cancel to leave 6 = 7. That is false, so there are **no solutions** — the two sides never meet.
P7	Very hard	If 3x − 4 = 11, what is the value of 6x + 2? 	32	3x = 15, so x = 5 and 6x = 30. 6x + 2 = **32**. A common wrong answer is x itself, and another is 6x — both are steps along the way, not the answer.
P8	Very hard	In the equation ax + 12 = 4x + b, a and b are constants. For which values does the equation have infinitely many solutions? 	a = 4 and b = 12	Infinitely many solutions means the two sides are the **same expression**, so the x coefficients match and the constants match. **a = 4 and b = 12.** A common wrong answer gives no solution; another gives exactly one.
P9	Very hard	Solve the equation ax = bx + c for x, where a ≠ b. 	c/(a − b)	x appears on both sides, so gather and factor: ax − bx = c gives x(a − b) = c. x = **c/(a − b)**. Dividing before factoring is what leads to the wrong answer.

#	Level	Question	Answer	Solution
P10	Very hard	A taxi charges a fixed fee plus a rate per kilometre. A 5 km ride costs \$11 and a 12 km ride costs \$25. What is the fixed fee, in dollars? 	1	The rate is (25 − 11)/(12 − 5) = 14/7 = \$2 per km. At 5 km: 11 = f + 2(5), so f = **\$1**.
P11	Very hard	If 2(3x − 1) = 4(x + 5) − 2, what is the value of x? 	10	6x − 2 = 4x + 20 − 2 = 4x + 18, so 2x = 20. x = **10**. Forgetting to distribute the 4 over both terms gives 8.
P12	Very hard	A student solves 4x + 7 = 2x − 5 and obtains x = 6. Substituting x = 6 gives 31 on the left and 7 on the right. What is the correct value of x? 	−6	Collecting correctly: 4x − 2x = −5 − 7, so 2x = −12. x = **−6**. Check: 4(−6) + 7 = −17 and 2(−6) − 5 = −17. The sign of the constant was the error.

The one habit to build. Substitute your answer back into the original equation. It takes ten seconds and catches almost every sign slip.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.

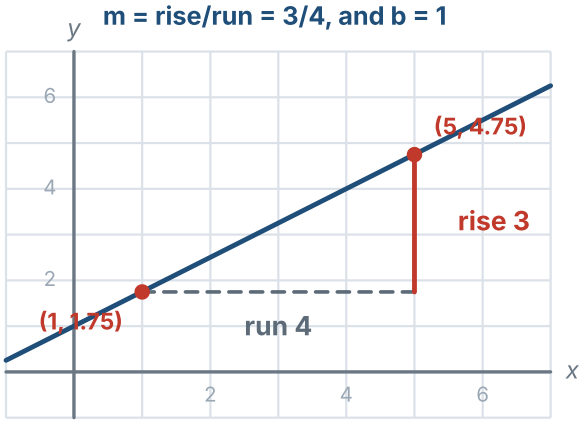
SAT Math 1.2 — Linear Functions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Slope as a rate of change

$$m = (y_2 - y_1)/(x_2 - x_1) = \text{rise over run}$$

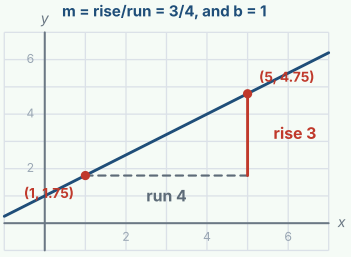
Slope is how much y changes for a one-unit rise in x. In a real context it always carries units: dollars per kilometre, degrees per hour. Keep the two points in the same order top and bottom.



From (1, 1.75) to (5, 4.75) the line climbs 3 while running 4, so the slope is 3/4. Swapping both differences gives −3/−4 — the same number.

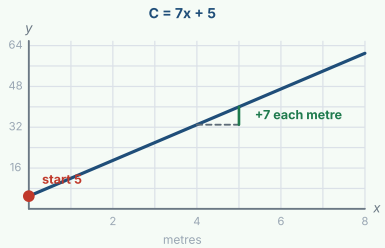
Example 1. **Find the slope through (1, 1.75) and (5, 4.75).**

Solution.
(4.75 − 1.75)/(5 − 1) = 3/4.
m = **0.75**.



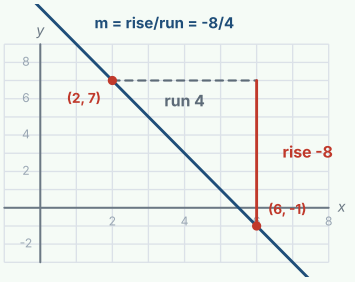
Example 2. **A tailor charges $19 for 2 m and $40 for 5 m. Find the rate per metre.**

Solution.
(40 − 19)/(5 − 2) = 21/3 = **$7 per metre**.



Example 3. **What is the slope of the line through (2, 7) and (6, −1)?**

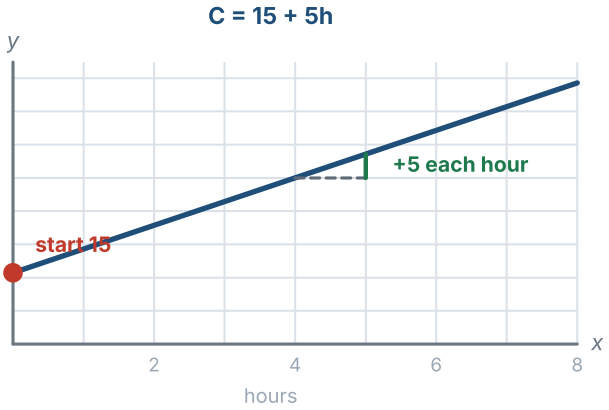
Solution.
(−1 − 7)/(6 − 2) = −8/4 = **−2**.



2 y = mx + b in context

b is the starting or fixed amount | m is the amount per unit

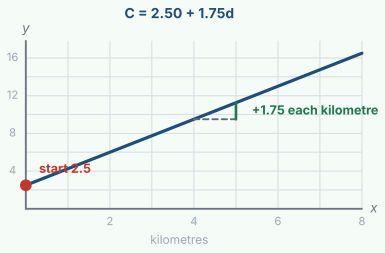
Interpretation questions are worth easy marks if you name the units. In C = 2.50 + 1.75d, the 2.50 is dollars and the 1.75 is dollars *per kilometre* — the 'per' is what makes it a rate.



The line starts at the fixed 15 and climbs 5 for every extra hour. Read b where the line meets the axis and m from one step across.

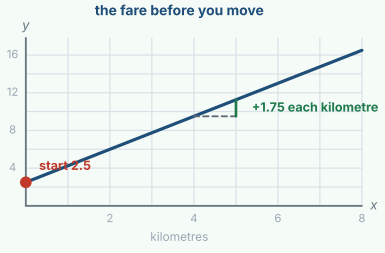
Example 1. **C = 2.50 + 1.75d gives a taxi fare for d kilometres. What does 1.75 mean?**

Solution.
The cost **per kilometre**, in dollars.



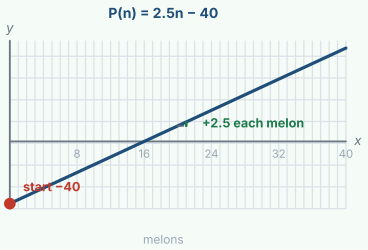
Example 2. **What does 2.50 mean in that model?**

Solution.
The **fixed starting fare** before any distance is travelled.



Example 3. **P(n) = 2.5n − 40 gives daily profit for n melons. What does 2.5 mean?**

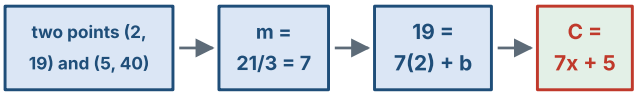
Solution.
The **increase in profit for each extra melon sold**, in dollars.



3 Writing a line from information

from two points: find m, then use y = mx + b with one point to get b

Three routes to the same line: two points, a point and a slope, or a table. Find the slope first in every case, then pin down b by substituting a known point.

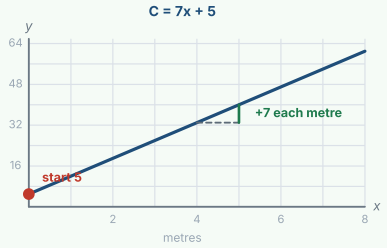


slope first, intercept second — always in that order

The rate comes from the two points; the fixed part comes from putting one of them back in. Doing it the other way round has nothing to substitute.

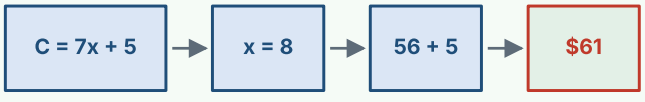
Example 1. **A tailor charges $19 for 2 m and $40 for 5 m. Write the model.**

Solution.
m = 7, and 19 = 7(2) + b gives b = 5.
C = 7x + 5.



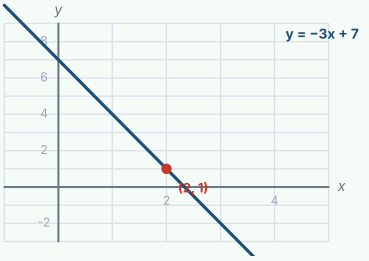
Example 2. **For that model, what is the charge for 8 metres?**

Solution.
7(8) + 5 = **$61**.



Example 3. **Write the equation of the line with slope −3 through (2, 1).**

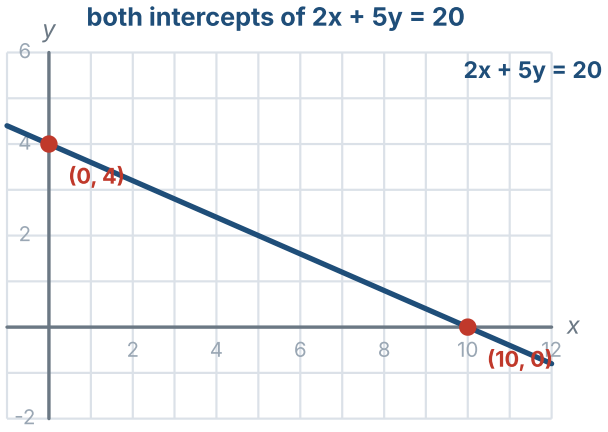
Solution.
1 = −3(2) + b gives b = 7.
y = −3x + 7.



4 Intercepts and what they mean

y-intercept: set x = 0 | x-intercept: set y = 0

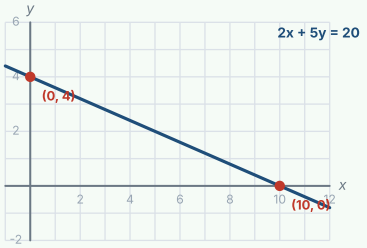
In context the y-intercept is the starting value and the x-intercept is when the quantity reaches zero — the break-even point, or when a tank empties.



Setting x = 0 gives the height where the line starts; setting y = 0 gives where it reaches the axis. Each is one substitution.

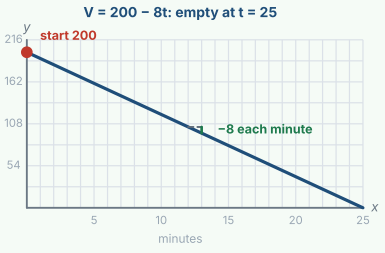
Example 1. **Find both intercepts of 2x + 5y = 20.**

Solution.
x = 0 gives y = 4; y = 0 gives x = 10.
(0, 4) and (10, 0).



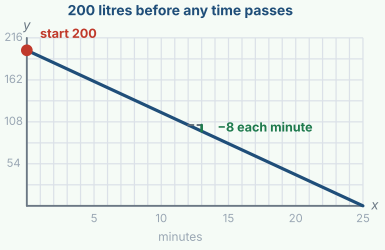
Example 2. **A tank model is V = 200 − 8t litres. When is it empty?**

Solution.
200 − 8t = 0, so t = **25 minutes**.



Example 3. **What does the 200 represent?**

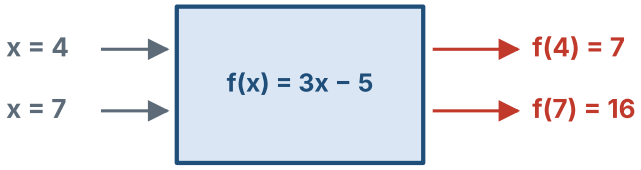
Solution.
The **starting volume**, in litres, at t = 0.



5 Function notation

f(3) means substitute 3 | f(x) = 3 means solve for x

The notation is compact but precise. Reading f(a) = b off a graph means the point (a, b) lies on it. Evaluating and solving are opposite directions, and the SAT tests both.

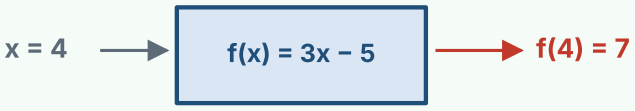


put the number in, read the number out

Put 4 in and 7 comes out, so f(4) = 7. Running it backwards from 16 asks which input produces that output.

Example 1. **f(x) = 3x − 5. Find f(4).**

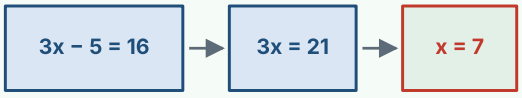
Solution.
3(4) − 5 = **7**.



put the number in, read the number out

Example 2. **For that f, solve f(x) = 16.**

Solution.
3x − 5 = 16, so 3x = 21.
x = **7**.



Example 3. **The graph of f passes through (2, 9). Write that in function notation.**

Solution.
f(2) = 9.

the point (2, 9)	is on the graph
in function notation	f(2) = 9

6 Linear models and tables

a constant difference in a table means the model is linear

Check the y-differences for equal steps. Once you know it is linear, the common difference is the slope and the value at x = 0 is b.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

Equal jumps of 3 for equal steps of 1: that is a slope of 3, and the value at x = 0 is the intercept.

Example 1. A table gives y = 5, 8, 11, 14 for x = 0, 1, 2, 3. Write the model.

Solution.
Differences are all 3, and y = 5 at x = 0.
y = 3x + 5.

	x = 0	x = 1	x = 2	x = 3
	5	8	11	14
differences		+3	+3	+3
ratios		×1.6	×1.38	×1.27

equal differences → linear

Example 2. A table gives y = 20, 17, 14 for x = 1, 2, 3. Write the model.

Solution.
Difference −3, and at x = 1, y = 20, so b = 23.
y = −3x + 23.

	x = 1	x = 2	x = 3
	20	17	14
differences		−3	−3
ratios		×0.85	×0.82

equal differences → linear

Example 3. **Is y = 2, 4, 8, 16 linear?**

Solution.

Differences 2, 4, 8 are not constant.

No — it is exponential, with ratio 2.

	x = 0	x = 1	x = 2	x = 3
	2	4	8	16
differences		+2	+4	+8
ratios		×2	×2	×2

equal ratios → exponential

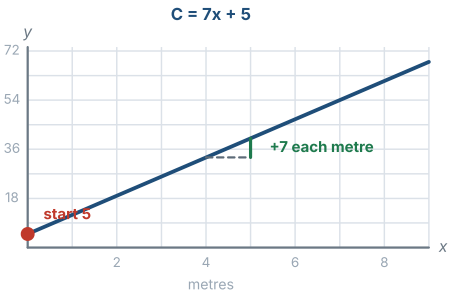
Part 1 — With Desmos (10 questions · every one is faster on the graph)

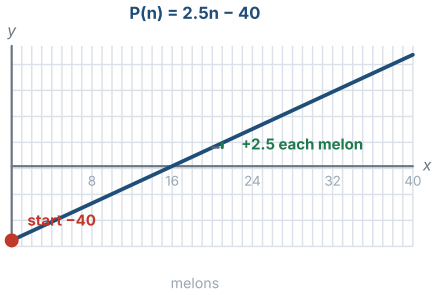
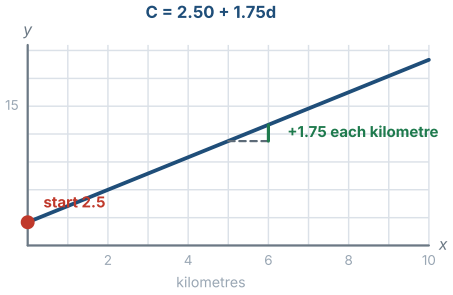
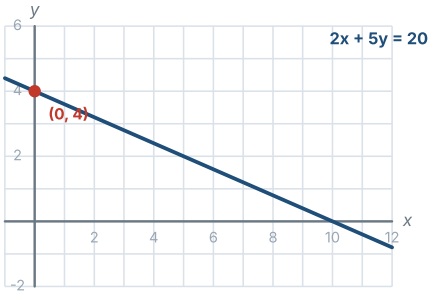
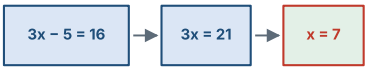
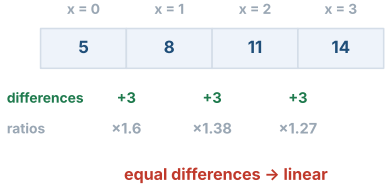
The SAT gives you Desmos on every maths question. Here it is **a line reader**. Plot two points, draw the line and Desmos gives the equation, the slope and both intercepts.

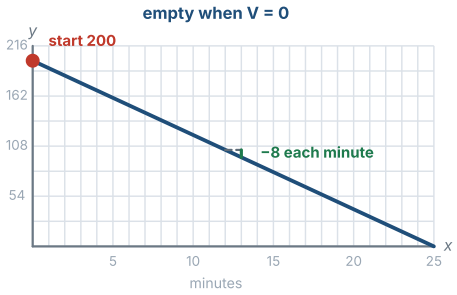
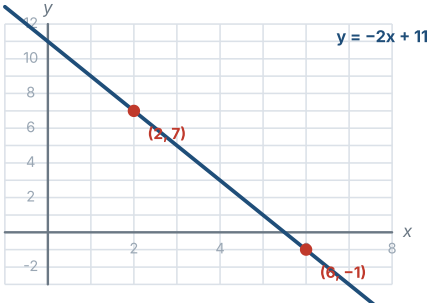
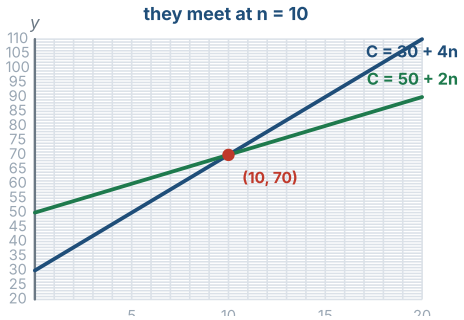
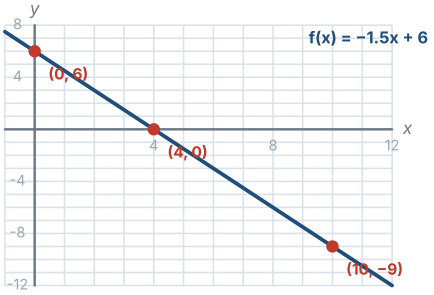
#	Question	Type into Desmos	Then	Answer
D1	What is the slope of the line through (1, 1.75) and (5, 4.75)?	`(4.75-1.75)/(5-1)`	Rise over run.	0.75
D2	A tailor charges $19 for 2 m and $40 for 5 m. What is the rate per metre?	`(40-19)/(5-2)`	The slope of the cost model.	7
D3	For that tailor, what is the charge for 8 metres?	`7*8+5`	Rate times amount plus the fixed part.	61
D4	What is the slope through (2, 7) and (6, −1)?	`(-1-7)/(6-2)`	Keep the order consistent.	-2
D5	Find the y-intercept of 2x + 5y = 20.	`20/5`	Set x = 0.	4
D6	Find the x-intercept of 2x + 5y = 20.	`20/2`	Set y = 0.	10
D7	V = 200 − 8t. At what t is V = 0?	`200/8`	The x-intercept in context.	25
D8	If f(x) = 3x − 5, what is f(4)?	`3*4-5`	Substitute.	7
D9	If f(x) = 3x − 5 and f(x) = 16, what is x?	`(16+5)/3`	Solve, do not substitute.	7
D10	A table gives y = 5, 8, 11, 14 at x = 0, 1, 2, 3. What is the slope?	`8-5`	Constant difference means linear.	3

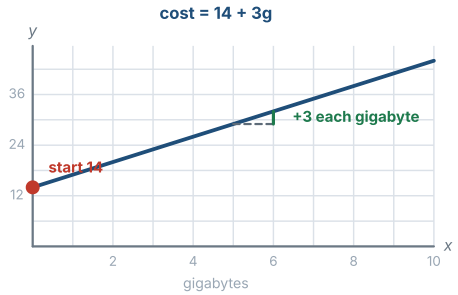
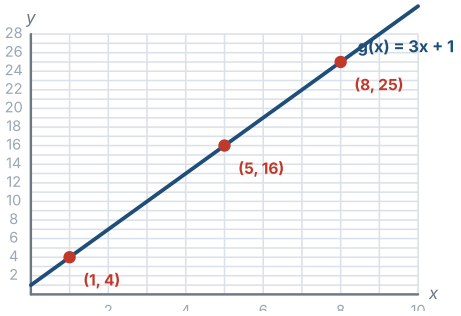
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Every interpretation question is answered by naming the **units**: dollars, or dollars per kilometre.

#	Level	Question	Answer	Solution
P1	Hard	A tailor's total charge is a linear function of the length of fabric used. The charge is $19 for 2 metres and $40 for 5 metres. What is the charge for 8 metres of fabric? 	$61	Rate = (40 − 19)/(5 − 2) = $7 per metre, and 19 = 7(2) + b gives b = $5. 7(8) + 5 = **$61**. A common wrong answer is 40 + 7×4, counting 5 to 8 as four steps instead of three; that answer drops the fixed $5. Only two of your thirteen got this on Test 5.

#	Level	Question	Answer	Solution
P2	Hard	The function P(n) = 2.5n − 40 gives the daily profit P, in dollars, of a fruit stall that sells n melons. What is the best interpretation of the number 2.5? 	**The increase in profit, in dollars, for each additional melon sold.**	The coefficient of n is a **rate**: profit per extra melon. A common wrong answer describes −40, the value when n = 0.
P3	Hard	The taxi fare, in dollars, for a ride of d kilometres is C = 2.50 + 1.75d. What is the best interpretation of 1.75? 	**The cost, in dollars, for each kilometre travelled.**	1.75 multiplies d, so it is the cost **per kilometre**. A common wrong answer is close but wrong: a 1 km ride costs 2.50 + 1.75 = $4.25.
P4	Hard	What is the y-coordinate of the y-intercept of the line 2x + 5y = 20? 	**4**	Set x = 0: 5y = 20. y = **4**.
P5	Hard	If f(x) = 3x − 5, for what value of x does f(x) = 16? 	**7**	3x − 5 = 16 gives 3x = 21. x = **7**. Substituting 16 instead would give 43, which answers a different question.
P6	Hard	A table shows y = 5, 8, 11 and 14 when x = 0, 1, 2 and 3. Write the equation that models the data. 	**y = 3x + 5**	The differences are all 3, so the slope is 3, and y = 5 when x = 0. **y = 3x + 5.** A common wrong answer swaps the slope and the intercept.

#	Level	Question	Answer	Solution
P7	Very hard	A water tank is modelled by V = 200 − 8t, where V is litres and t is minutes. After how many minutes is the tank empty? 	25	Set V = 0: 200 = 8t. t = **25 minutes**. This is the x-intercept read in context.
P8	Very hard	A line passes through (2, 7) and (6, −1). Write the equation of that line. 	y = −2x + 11	Slope = −8/4 = −2, and 7 = −2(2) + b gives b = 11. **y = −2x + 11.** A common wrong answer uses the y-value of the first point as the intercept.
P9	Very hard	Two linear models for a monthly cost are C = 30 + 4n and C = 50 + 2n, where n is the number of items. For what value of n do the two cost the same? 	10	30 + 4n = 50 + 2n gives 2n = 20. n = **10 items**, where both cost $70.
P10	Very hard	The graph of a linear function f passes through (0, 6) and (4, 0). What is the value of f(10)? 	−9	Slope = (0 − 6)/(4 − 0) = −1.5, and b = 6, so f(x) = −1.5x + 6. f(10) = −15 + 6 = **−9**. A common wrong answer assumes the line stops at zero.

#	Level	Question	Answer	Solution
P11	Very hard	A phone plan charges a fixed monthly fee plus a rate per gigabyte. Using 4 GB costs $26 and using 9 GB costs $41. What is the fixed monthly fee, in dollars? 	14	Rate = (41 − 26)/(9 − 4) = 15/5 = $3 per GB. 26 = f + 3(4), so f = **$14**. A common wrong answer is the rate, not the fee.
P12	Very hard	For a linear function g, g(1) = 4 and g(5) = 16. What is the value of g(8)? 	25	Slope = (16 − 4)/(5 − 1) = 3, and 4 = 3(1) + b gives b = 1. g(8) = 24 + 1 = **25**. A common wrong answer adds the slope once rather than three times.

The one habit to build. Find the rate first and the fixed part second. On Test 5 only two of thirteen got the tailor question, and the usual cause was counting the step from 5 to 8 as four instead of three.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.

SAT Math 1.3 — Linear Equations in Two Variables

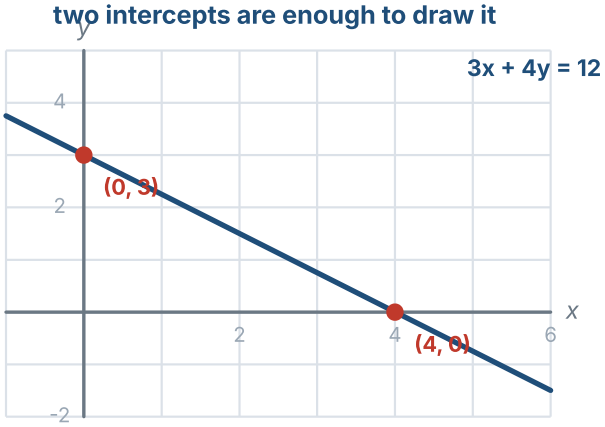
Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Graphing a line

plot both intercepts and join them | or start at b and step by the slope

From standard form, the intercepts are the fastest two points: set x = 0, then set y = 0. From slope-intercept form, plot b on the y-axis and step rise over run.

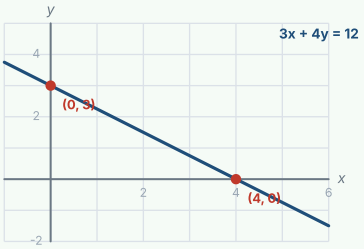
two intercepts are enough to draw it



Two points fix a line, and the intercepts are the two easiest to find — each costs one substitution.

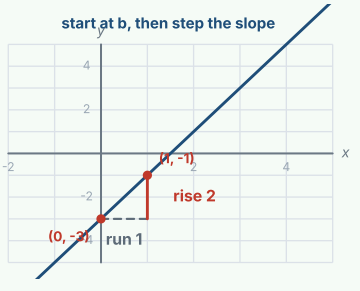
Example 1. **Find both intercepts of 3x + 4y = 12.**

Solution.
x = 0 gives y = 3; y = 0 gives x = 4.
(0, 3) and (4, 0).



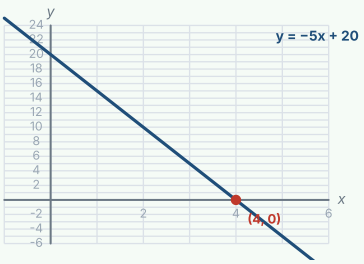
Example 2. **Describe how to graph y = 2x − 3 without a table.**

Solution.
Start at **(0, −3)**, then go up 2 and right 1 to reach (1, −1).



Example 3. **Find the x-intercept of y = −5x + 20.**

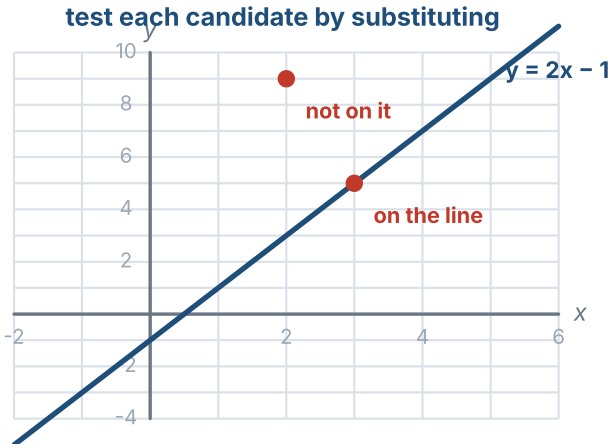
Solution.
0 = −5x + 20, so x = **4**.



2 Is this point on the line?

substitute both coordinates and see whether the equation holds

There is nothing clever here, and that is the point: a question asking which point lies on a line is answered by testing options until one works.



A point is on the line exactly when its coordinates satisfy the equation. One substitution settles it.

Example 1. **Does (3, 5) lie on y = 2x − 1?**

Solution.
2(3) − 1 = 5, which matches.
Yes.

substitute x and compare with y

(3, 5)	2(3) − 1 = 5 ✓	✓ keep

Example 2. **Does (2, 9) lie on 4x + y = 16?**

Solution.
8 + 9 = 17, not 16.
No.

substitute x and y into the equation

(2, 9)	8 + 9 = 17, not 16	× reject

Example 3. **The point (k, 7) lies on y = 3x + 1. Find k.**

Solution.
7 = 3k + 1, so 3k = 6.
k = **2**.

7 = 3k + 1 → 3k = 6 → k = 2

3 The same line in different clothes

slope-intercept $y = mx + b$ | standard $Ax + By = C$ | point-slope $y - y_1 = m(x - x_1)$

Two equations describe the same line when one is a multiple of the other. Rearranging to $y = mx + b$ is the quickest way to compare.

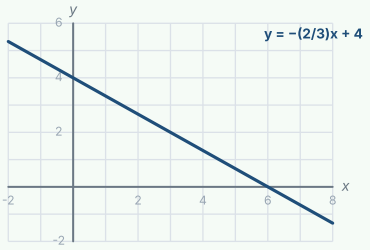
one line, three appearances

$2x + 3y = 12$	standard form
$y = -(2/3)x + 4$	the same line, rearranged
$4x + 6y = 24$	the same line again, doubled

Doubling every coefficient changes nothing about the line. Putting each into $y = mx + b$ makes that obvious at a glance.

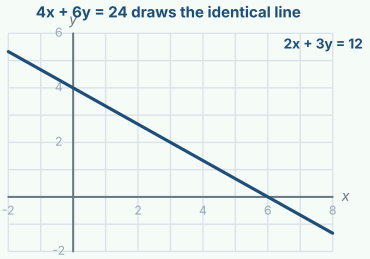
Example 1. **Write $2x + 3y = 12$ in slope-intercept form.**

Solution.
$3y = -2x + 12$.
$y = -(2/3)x + 4$.



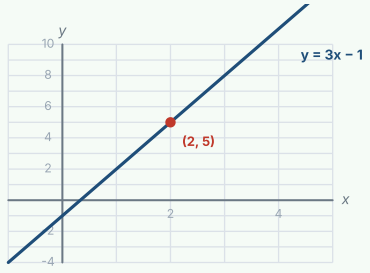
Example 2. **Are $2x + 3y = 12$ and $4x + 6y = 24$ the same line?**

Solution.
The second is exactly twice the first.
Yes — one line, written two ways.



Example 3. **Write the line through (2, 5) with slope 3 in point-slope form.**

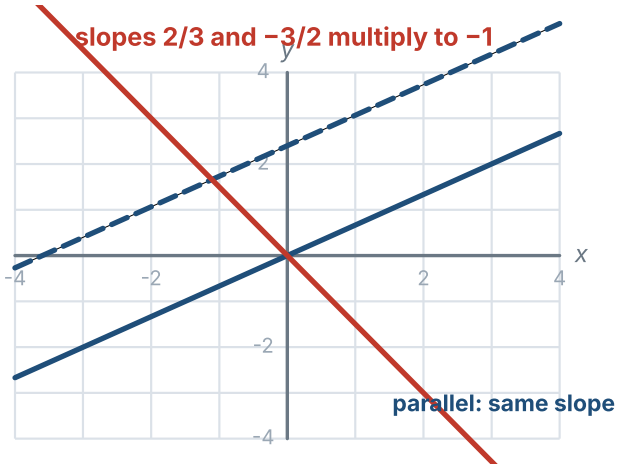
Solution.
$y - 5 = 3(x - 2)$, which rearranges to $y = 3x - 1$.



4 Parallel and perpendicular

parallel → same slope | perpendicular → slopes multiply to −1

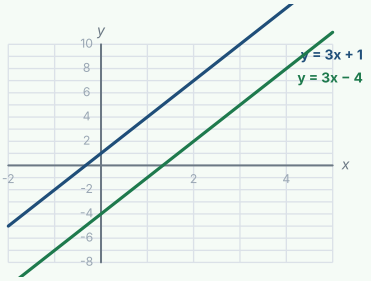
Rearrange both equations into y = mx + b before comparing. To get a perpendicular slope, flip the fraction and change the sign.



The two blue lines share a slope and never meet; the red one crosses at a right angle, with the slope flipped and negated.

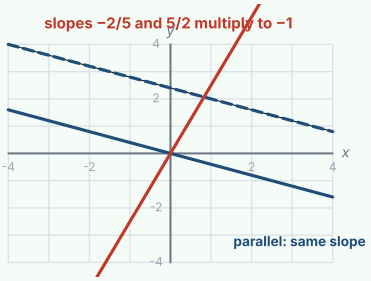
Example 1. **Is y = 3x + 1 parallel to 6x − 2y = 8?**

Solution.
The second gives y = 3x − 4, the same slope.
Yes, parallel.



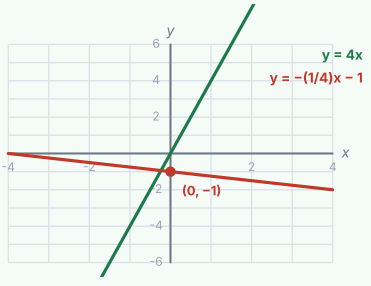
Example 2. **Find the slope perpendicular to y = −(2/5)x + 1.**

Solution.
Flip and negate.
5/2.



Example 3. **Write the line through (0, −1) perpendicular to y = 4x.**

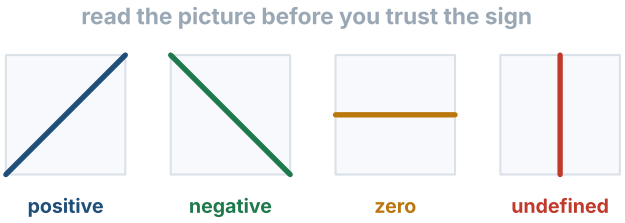
Solution.
Perpendicular slope −1/4.
y = −(1/4)x − 1.



5 Horizontal and vertical lines

y = c is horizontal, slope 0 | x = c is vertical, slope undefined

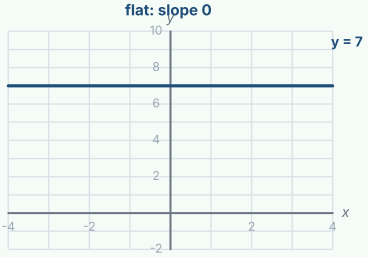
A horizontal line has zero slope because nothing rises. A vertical line has no slope at all, because the run is zero and you cannot divide by it — 'zero slope' and 'undefined slope' are different things.



Four lines, four slopes: positive, negative, zero (flat) and undefined (vertical). The last two are the pair that get confused.

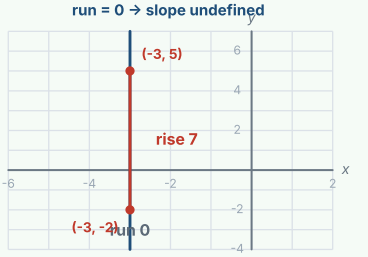
Example 1. **What is the slope of y = 7?**

Solution.
0 — it is horizontal.



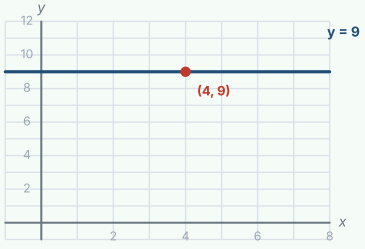
Example 2. **What is the slope of x = −3?**

Solution.
Undefined — it is vertical, and the run is zero.



Example 3. **Write the line through (4, 9) that is parallel to the x-axis.**

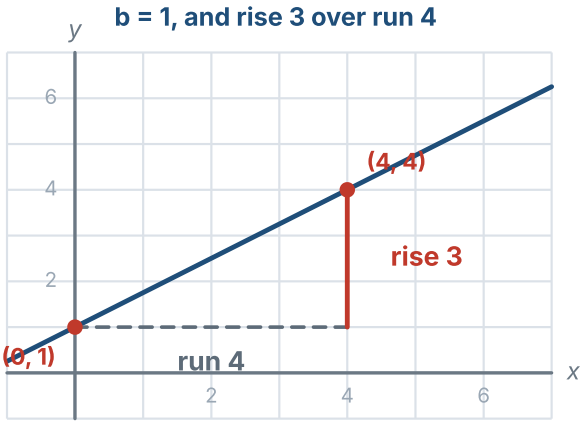
Solution.
Parallel to the x-axis means horizontal.
y = 9.



6 Reading a line off a graph

find b where it crosses the y-axis, then count rise over run between two grid points

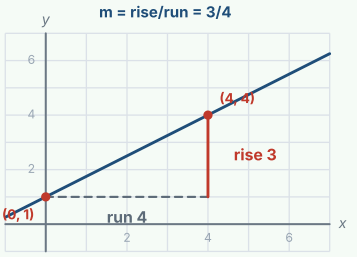
Pick two points where the line passes exactly through grid intersections, or the slope you count will be wrong. Then read b directly.



Both marked points sit on grid corners, so the count is exact: up 3, across 4, starting from a y-intercept of 1.

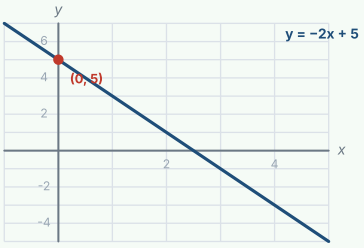
Example 1. **A line crosses the y-axis at 1 and passes through (4, 4). Find its equation.**

Solution.
m = (4 − 1)/4 = 3/4.
y = (3/4)x + 1.



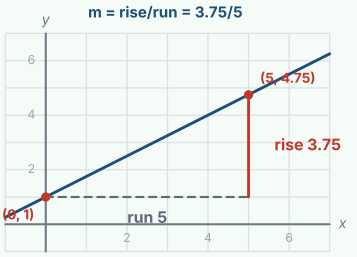
Example 2. **A line falls 2 for every 1 across and passes through (0, 5). Find its equation.**

Solution.
y = −2x + 5.



Example 3. **A graph shows a line through (0, 1) and (5, 4.75). Find its slope.**

Solution.
(4.75 − 1)/5 = **0.75**.



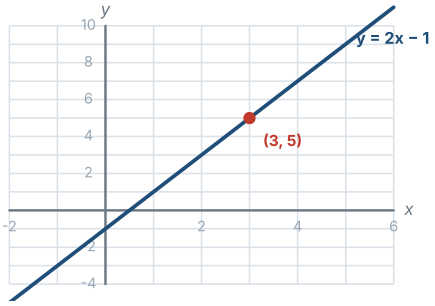
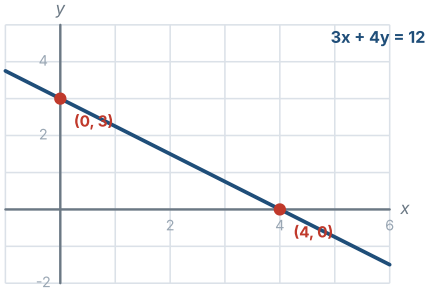
Part 1 — With Desmos (10 questions · every one is faster on the graph)

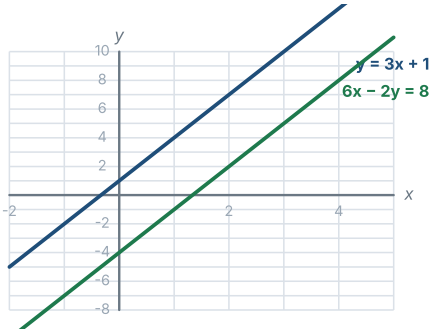
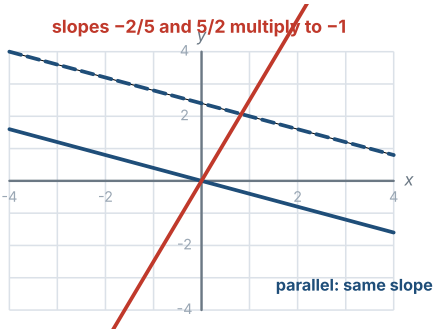
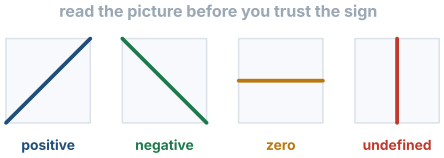
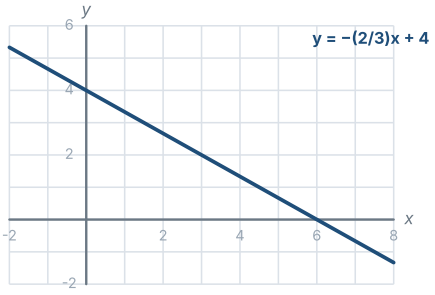
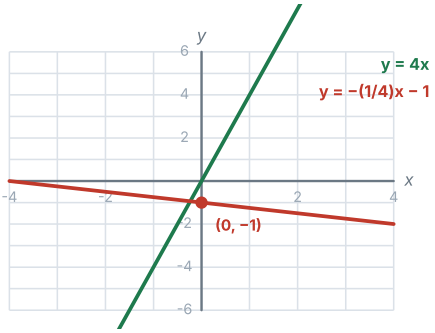
The SAT gives you Desmos on every maths question. Here it is **a grapher**. Type any form of the equation and read the slope and the intercepts straight off the picture.

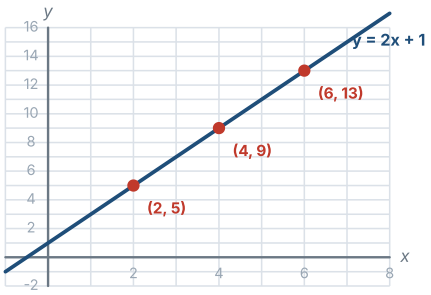
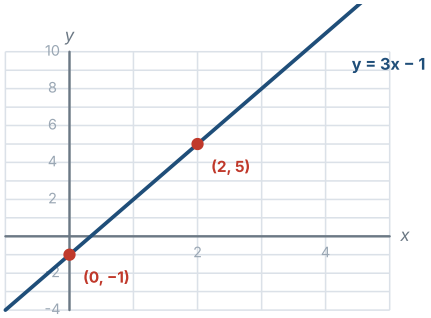
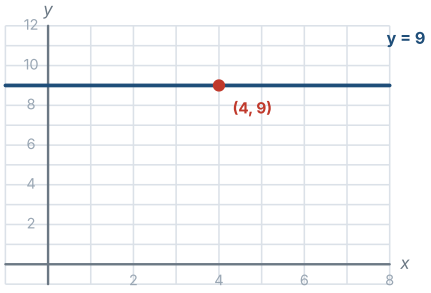
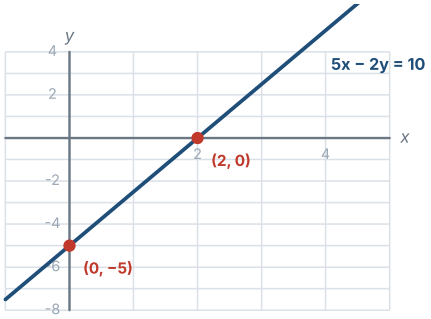
#	Question	Type into Desmos	Then	Answer
D1	What is the y-intercept of 3x + 4y = 12?	`12/4`	Set x = 0.	3
D2	What is the x-intercept of 3x + 4y = 12?	`12/3`	Set y = 0.	4
D3	Does the point (3, 5) lie on y = 2x − 1?	`2*3-1`	Compare with 5.	yes
D4	The point (k, 7) is on y = 3x + 1. What is k?	`(7-1)/3`	Solve for the input.	2
D5	Write 2x + 3y = 12 in the form y = mx + b. What is m?	`-2/3`	Rearranging is easier than it looks.	-2/3
D6	What slope is perpendicular to −2/5?	`-1/(-2/5)`	Flip and negate.	2.5
D7	Are y = 3x + 1 and 6x − 2y = 8 parallel?	`6/2`	Rearrange the second to compare.	yes, both slope 3
D8	What is the slope of the line through (0, 1) and (5, 4.75)?	`(4.75-1)/5`	Rise over run.	0.75
D9	What is the x-intercept of y = −5x + 20?	`20/5`	Set y = 0.	4
D10	A line has slope −2 and passes through (0, 5). What is y at x = 3?	`-2*3+5`	Substitute into y = mx + b.	-1

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Rearrange to y = mx + b before comparing two lines — two equations can look different and be the same line.

#	Level	Question	Answer	Solution
P1	Hard	Give a point that lies on the line y = 2x − 1. 	(3, 5)	Test each: at x = 3, y = 2(3) − 1 = **5**. A common wrong answer gives 2(2) − 1 = 3, not 5 — close enough to catch a careless reader.
P2	Hard	What is the x-intercept of the line 3x + 4y = 12? 	4	Set y = 0: 3x = 12. x = **4**. Setting x = 0 instead gives the y-intercept, 3.

#	Level	Question	Answer	Solution
P3	Hard	Write the equation of a line parallel to y = 3x + 1. 	6x − 2y = 8	Rearranging 6x − 2y = 8 gives y = 3x − 4, which has the same slope. **Parallel.** A common wrong answer negates the slope, and another inverts it — neither is parallel.
P4	Hard	A line has slope −2/5. What is the slope of a line perpendicular to it? 	5/2	Perpendicular slopes are negative reciprocals: flip and change the sign. **5/2.** A common wrong answer flips without negating.
P5	Hard	What is the slope of the line x = −3? 	undefined	A vertical line has a run of zero, and division by zero is undefined. **Undefined.** A common wrong answer is the slope of a *horizontal* line — the two are often confused.
P6	Hard	Write the equation that 2x + 3y = 12 becomes in slope-intercept form. 	y = −(2/3)x + 4	3y = −2x + 12, then divide by 3. **y = −(2/3)x + 4.** A common wrong answer inverts the slope.
P7	Very hard	Line k passes through (0, −1) and is perpendicular to the line y = 4x. What is the equation of line k? 	y = −(1/4)x − 1	The perpendicular slope to 4 is −1/4, and the y-intercept is given as −1. **y = −(1/4)x − 1.** A common wrong answer negates without flipping.

#	Level	Question	Answer	Solution
P8	Very hard	A line passes through (2, 5) and (6, 13). Give one other point with integer coordinates that lies on this line. 	(4, 9)	Slope = (13 − 5)/(6 − 2) = 2, and 5 = 2(2) + b gives b = 1, so y = 2x + 1. At x = 4: y = **9**, so (4, 9). A common wrong answer would need y = 21.
P9	Very hard	The equations 2x + 3y = 12 and 4x + 6y = k represent the same line. What is the value of k? 2x + 3y = 12 — double everything 4x + 6y = 24 — so k = 24	24	The second equation must be a multiple of the first; doubling 2x + 3y = 12 gives 4x + 6y = 24. k = **24**.
P10	Very hard	A line with slope 3 passes through the point (2, 5). What is its y-intercept? 	−1	5 = 3(2) + b gives b = 5 − 6. b = **−1**. A common wrong answer adds instead of subtracting.
P11	Very hard	Which is the equation of the line through (4, 9) that is parallel to the x-axis? 	y = 9	Parallel to the x-axis means **horizontal**, so y is constant at the given y-value. **y = 9.** A common wrong answer is the vertical line through that point.
P12	Very hard	In the xy-plane, line m has equation 5x − 2y = 10. What is the sum of its x-intercept and its y-intercept? 	−3	Set y = 0: x = 2. Set x = 0: −2y = 10, so y = −5. 2 + (−5) = **−3**.

The one habit to build. Zero slope and undefined slope are different things. A horizontal line has slope 0; a vertical line has none at all.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.

SAT Math 1.4 — Systems of Linear Equations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Elimination

line the equations up and add or subtract to kill one variable

Elimination is usually faster than substitution on the SAT, because the coefficients are chosen to cooperate. If they do not match, multiply one equation through first.

line them up, then add to kill one variable

$$\begin{array}{rr} & 2x + y = 11 \\ + & x - y = 4 \\ \hline & 3x = 15 \end{array}$$

$x = 5$, then $y = 1$

The y terms are +y and −y, so adding the two equations removes y outright and leaves one equation in x.

Example 1. **Solve $2x + y = 11$ and $x - y = 4$.**

Solution.

Adding kills y: $3x = 15$, so $x = 5$.

Then $y = 1$, giving **(5, 1)**.

line them up, then add to kill one variable

$$\begin{array}{rr} & 2x + y = 11 \\ + & x - y = 4 \\ \hline & 3x = 15 \end{array}$$

$x = 5$, $y = 1$

Example 2. **Solve $3a + 2p = 8$ and $a + 2p = 6$.**

Solution.

Subtracting kills p: $2a = 2$, so $a = 1$.

Then $p = 2.5$, giving **$a = 1$, $p = 2.5$**.

line them up, then subtract to kill one variable

$$\begin{array}{rr} & 3a + 2p = 8 \\ - & a + 2p = 6 \\ \hline & 2a = 2 \end{array}$$

$a = 1$, $p = 2.5$

Example 3. **Solve 3x + 2y = 57 and x + 4y = 39.**

Solution.
Multiply the second by 3: 3x + 12y = 117. Subtract: 10y = 60, so y = 6.
Then x = 15, giving **(15, 6)**.

line them up, then subtract to kill one variable

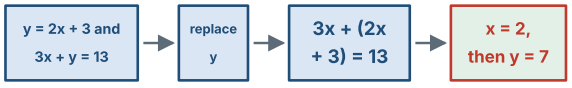
$$\begin{array}{rr} & 3x + 2y = 57 \\ - & 3x + 12y = 117 \\ \hline & 10y = 60 \end{array}$$

$$y = 6,\ x = 15$$

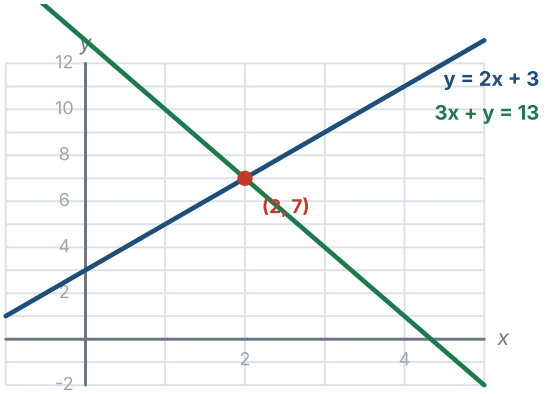
2 Substitution

when one variable is already alone, put that expression into the other equation

Substitution is the natural choice when an equation reads y = something. Replace, solve for the remaining variable, then go back for the other one.



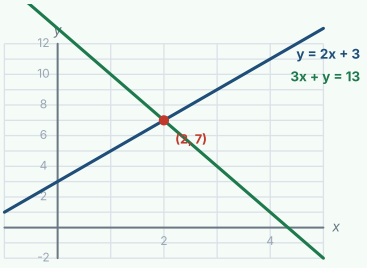
solve for one, then go back for the other



The algebra and the picture agree: the pair (2, 7) is the one point that satisfies both equations at once.

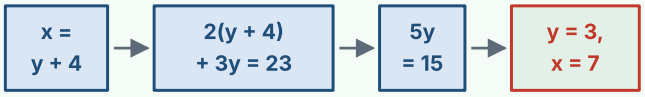
Example 1. **Solve y = 2x + 3 and 3x + y = 13.**

Solution.
3x + (2x + 3) = 13 gives 5x = 10, so x = 2.
Then y = 7: **(2, 7)**.



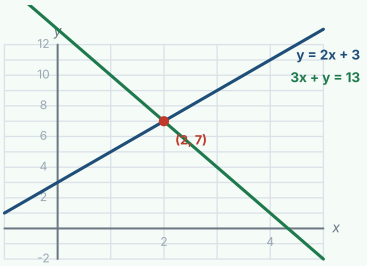
Example 2. **Solve x = y + 4 and 2x + 3y = 23.**

Solution.
2(y + 4) + 3y = 23 gives 5y = 15, so y = 3.
Then x = 7: **(7, 3)**.



Example 3. **Why finish by finding the second variable?**

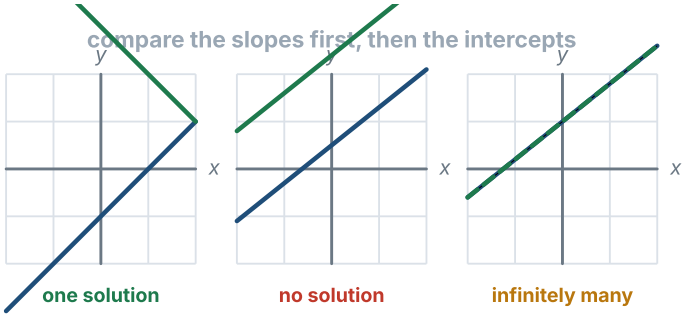
Solution.
A system's solution is a **point**. Stopping at x answers only half the question — though do check whether the question wanted only one of them.



3 One, none, or infinitely many

different slopes → one | same slope, different intercept → none | identical → infinitely many

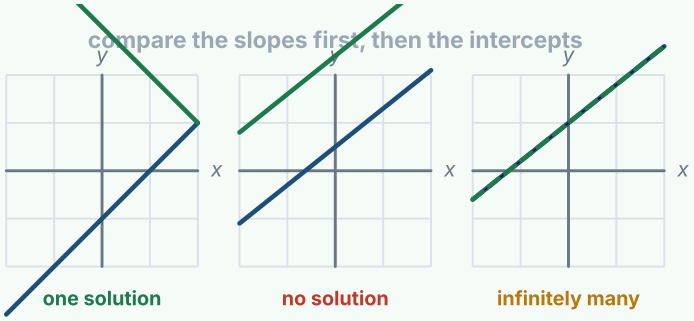
Two lines either cross once, run parallel forever, or sit on top of each other. For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, compare the ratios a_1/a_2, b_1/b_2 and c_1/c_2.



Crossing once, never meeting, or lying on top of one another — and the ratios of the coefficients tell you which before you solve anything.

Example 1. **How many solutions does 3x + 5y = 21 and 6x + 10y = 42 have?**

Solution.
The second is exactly twice the first.
Infinitely many — one line.

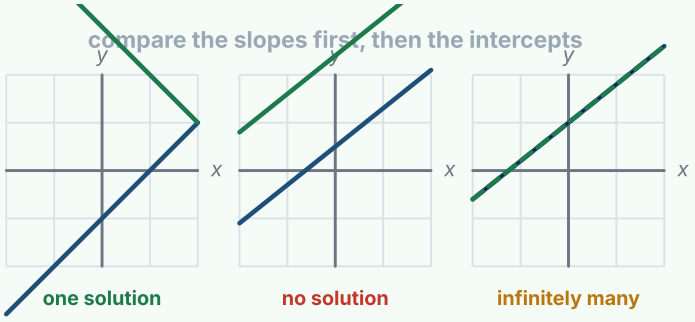


Example 2. **How many solutions does 3x + 5y = 21 and 6x + 10y = 35 have?**

Solution.

Same slope, different constant: parallel.

None.

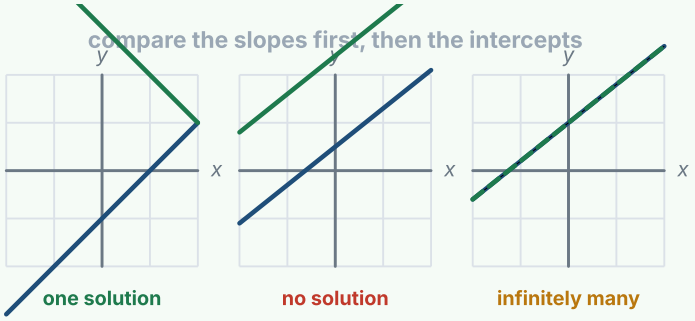


Example 3. **How many does 3x + 5y = 21 and 2x − y = 4 have?**

Solution.

The slopes differ.

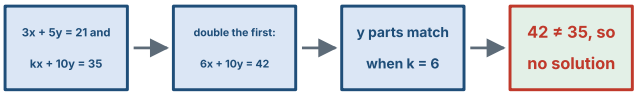
Exactly one.



4 Finding the parameter

no solution → match the coefficients but not the constants | infinitely many → match everything

These questions give you a letter in one coefficient and ask you to force a case. Scale one equation until the x and y parts line up, then compare the constants.

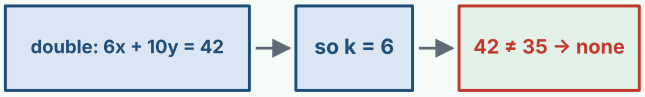


match the coefficients, then check the constants

Doubling makes the y terms agree, which fixes k. The constants then decide whether the answer is 'no solution' or 'infinitely many'.

Example 1. **For what k does 3x + 5y = 21 and kx + 10y = 35 have no solution?**

Solution.
Doubling the first gives 6x + 10y = 42, so k = **6**, and 42 ≠ 35 confirms no solution.



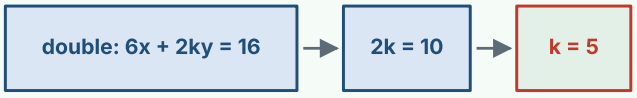
Example 2. **For what k does 2x + 6y = 10 and kx + 9y = 15 have infinitely many solutions?**

Solution.
Multiplying the first by 1.5 gives 3x + 9y = 15.
k = **3**.



Example 3. **For what value of k does 3x + ky = 8 and 6x + 10y = 9 have no solution?**

Solution.
Doubling the first gives 6x + 2ky = 16, so 2k = 10.
k = **5**.



5 Two-constraint word problems

one equation for the count, one for the value

Tickets, coins, mixtures and two-rate problems all share this shape: how many things, and how much they are worth. Define both variables before writing anything.

two sentences, two equations

how many: a + s = 45	the count equation
how much: 12a + 7s = 440	the value equation
substitute s = 45 − a	5a = 125, so a = 25

The first equation counts the tickets; the second prices them. Two unknowns need exactly these two facts.

Example 1. **Adult tickets cost $12 and student $7. 45 tickets sold for $440. How many adult tickets?**

Solution.
a + s = 45 and 12a + 7s = 440.
Substituting: 12a + 7(45 − a) = 440 gives 5a = 125, so a = **25**.

a + s = 45	count
12a + 7s = 440	value
5a = 125	a = 25

Example 2. **3 apples and 2 pears cost $8.00; 1 apple and 2 pears cost $6.00. Find the price of a pear.**

Solution.
Subtracting: 2 apples cost $2, so an apple is $1.
Then 2 pears cost $5: a pear is **$2.50**.

line them up, then subtract to kill one variable

$$\begin{array}{rr} & 3a + 2p = 8 \\ - & a + 2p = 6 \\ \hline & 2a = 2 \end{array}$$

a = 1, so p = 2.50

Example 3. **3 crates and 2 baskets cost 57; 1 crate and 4 baskets cost 39. Find the crate price.**

Solution.
Multiply the second by 3 and subtract: 10 baskets = 60, so a basket is 6.
A crate is (57 − 12)/3 = **15**.

line them up, then subtract to kill one variable

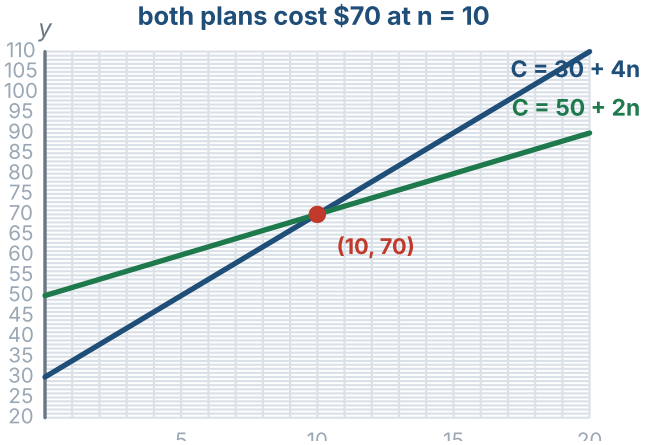
$$\begin{aligned} & 3c + 2b = 57 \\ - \quad & 3c + 12b = 117 \\ \hline & 10b = 60 \end{aligned}$$

$$b = 6,\ c = 15$$

6 Interpreting the solution

the solution is the point where both conditions hold at once

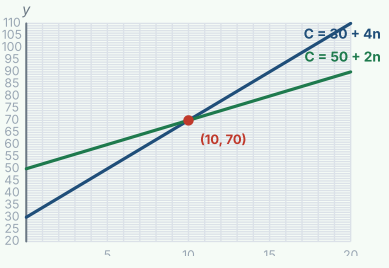
In context the intersection is the break-even point, the moment two plans cost the same, or the mix that satisfies both requirements. Check your answer makes sense: you cannot sell 2.5 tickets.



Before n = 10 one plan is cheaper; after it, the other. The crossing point is the only place they agree.

Example 1. **Two plans cost C = 30 + 4n and C = 50 + 2n. What does their intersection mean?**

Solution.
The number of items at which **both plans cost the same**: n = 10, at $70.



Example 2. **A system gives a = 25 and s = 20 for tickets. Check it against 45 tickets for $440.**

Solution.

25 + 20 = 45 ✓ and 12(25) + 7(20) = 300 + 140 = 440 ✓.

Both hold.

a solution must satisfy both equations

count: 25 + 20	= 45 ✓	✓ keep
value: 300 + 140	= 440 ✓	✓ keep

Example 3. **A ticket system solves to a = 22.5. What does that tell you?**

Solution.

That something is wrong — you cannot sell **half a ticket**. Recheck the setup.

does the answer make sense in context?

a = 22.5	tickets come in whole numbers	× reject

Part 1 — With Desmos (10 questions · every one is faster on the graph)

The SAT gives you Desmos on every maths question. Here it is **an intersection finder**. Enter both equations and click where the lines cross; Desmos gives the point exactly.

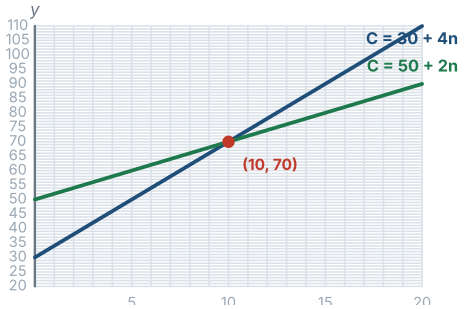
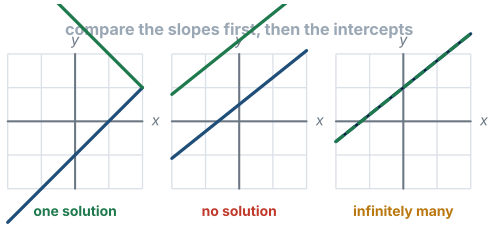
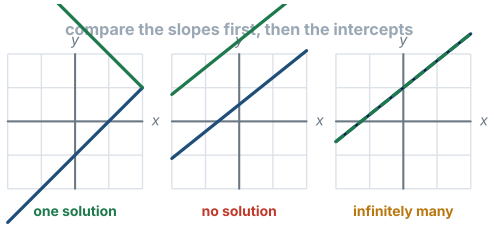
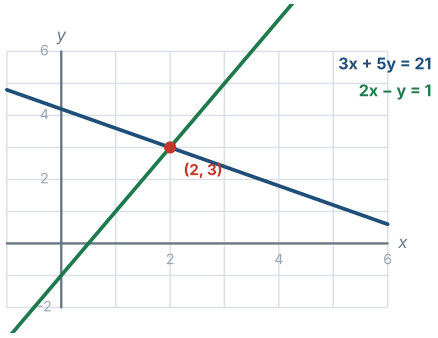
#	Question	Type into Desmos	Then	Answer
D1	Solve 2x + y = 11 and x − y = 4. What is x?	`2x+y=11, x-y=4`	Enter both equations and click the intersection.	5
D2	For that system, what is x + y?	`5+1`	Find both, then combine.	6
D3	Solve y = 2x + 3 and 3x + y = 13. What is x?	`y=2x+3, 3x+y=13`	Desmos marks the crossing point.	2
D4	3 apples and 2 pears cost $8.00; 1 apple and 2 pears cost $6.00. What does a pear cost?	`3a+2p=8, a+2p=6`	Two equations, two unknowns.	2.5
D5	Adult tickets cost $12 and student $7; 45 tickets for $440. How many adult tickets?	`a+s=45, 12a+7s=440`	Count equation and value equation.	25
D6	3 crates and 2 baskets cost 57; 1 crate and 4 baskets cost 39. What does a crate cost?	`3c+2b=57, c+4b=39`	Same structure again.	15
D7	For what k do 3x + ky = 8 and 6x + 10y = 9 have no solution?	`10/2`	Match the coefficients.	5
D8	For what k do 2x + 6y = 10 and kx + 9y = 15 have infinitely many solutions?	`2*1.5`	Scale the whole first equation.	3
D9	Two plans cost 30 + 4n and 50 + 2n. At what n are they equal?	`30+4n=50+2n`	Set them equal.	10
D10	Solve 3x + 2y = 57 and x + 4y = 39. What is y?	`3x+2y=57, x+4y=39`	Eliminate x first if you work by hand.	6

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. For word problems write one equation for the **count** and one for the **value** before solving anything.

#	Level	Question	Answer	Solution
P1	Hard	The system 2x + y = 11 and x − y = 4 has solution (x, y). What is the value of x + y? line them up, then add to kill one variable 2x + y = 11 + x − y = 4 3x = 15 x = 5, y = 1, so x + y = 6	6	Adding the equations gives 3x = 15, so x = 5, and then y = 1. x + y = **6**. A common wrong answer is x + 4, and another is the first constant — both are numbers already in the problem.
P2	Hard	At a farmers market, 3 apples and 2 pears cost $8.00, and 1 apple and 2 pears cost $6.00. What is the price of one pear? line them up, then subtract to kill one variable 3a + 2p = 8 − a + 2p = 6 2a = 2 a = $1, so p = $2.50	$2.50	Subtracting the equations removes the pears: 2 apples cost $2, so an apple is $1. Then 2 pears cost $5, so a pear is **$2.50**. A common wrong answer is the apple price.

#	Level	Question	Answer	Solution
P3	Hard	At a basketball tournament, adult tickets cost $12 each and student tickets cost $7 each. A total of 45 tickets were sold for $440. How many adult tickets were sold? a + s = 45 — the count 12a + 7s = 440 — the value 5a = 125 — a = 25 adult tickets	25	a + s = 45 and 12a + 7s = 440. Substituting s = 45 − a gives 12a + 315 − 7a = 440. 5a = 125, so a = **25**.
P4	Hard	In the system 3x + ky = 8 and 6x + 10y = 9, k is a constant. For which value of k does the system have no solution? double: 6x + 2ky = 16 → 2k = 10 → k = 5 → 16 ≠ 9 → none	5	No solution means the lines are parallel, so doubling the first must match the second's x and y parts: 6x + 2ky = 16 needs 2k = 10. k = **5**, and 16 ≠ 9 confirms they never meet.
P5	Hard	At a market, 3 crates of oranges and 2 baskets of figs cost 57 talens, while 1 crate and 4 baskets cost 39 talens. What is the price, in talens, of one crate of oranges? line them up, then subtract to kill one variable 3c + 2b = 57 − 3c + 12b = 117 10b = 60 b = 6, so c = 15	15	Multiply the second equation by 3 and subtract: 10 baskets cost 60, so a basket is 6. Then 3 crates cost 57 − 12 = 45, so a crate is **15**. A common wrong answer is the basket price.
P6	Hard	How many solutions does the system 3x + 5y = 21 and 6x + 10y = 35 have? compare the slopes first, then the intercepts one solution — no solution — infinitely many	Zero	Doubling the first gives 6x + 10y = 42, which contradicts 35. Same slope, different intercept means the lines are **parallel** and never meet.
P7	Very hard	A weaver's pricing system is modelled by 2x + 6y = 10 and kx + 9y = 15, where k is a constant. If the system has infinitely many solutions, what is the value of k? 9/6 = 1.5 → 15/10 = 1.5 ✓ → k = 2 × 1.5 → k = 3	3	Infinitely many solutions means one equation is a multiple of the other. The y terms give the factor: 9/6 = 1.5, and 15/10 = 1.5 agrees. k = 2 × 1.5 = **3**.

#	Level	Question	Answer	Solution
P8	Very hard	Two mobile plans charge C = 30 + 4n and C = 50 + 2n dollars for n items. At the number of items where the costs are equal, what is that cost, in dollars? 	70	30 + 4n = 50 + 2n gives 2n = 20, so n = 10. C = 30 + 40 = **$70**. Answering 10 gives the number of items, not the cost.
P9	Very hard	A chemist mixes a 20% solution with a 50% solution to make 30 litres of a 40% solution. How many litres of the 20% solution are used? x + y = 30 — the volume equation 0.2x + 0.5y = 12 — the concentration equation −0.3x = −3 — x = 10 litres	10	Let x be the 20% volume: x + y = 30 and 0.2x + 0.5y = 12. Substituting: 0.2x + 0.5(30 − x) = 12 gives −0.3x = −3, so x = **10 litres**.
P10	Very hard	The system y = 2x + 3 and y = ax + 7 has no solution. What is the value of a? 	2	No solution means the lines are parallel, so the slopes match. a = **2**, and because 3 ≠ 7 the lines really are distinct.
P11	Very hard	A system of two linear equations is graphed, and the two lines lie exactly on top of one another. State what is true here. 	The system has infinitely many solutions.	Every point on the shared line satisfies both equations. **Infinitely many solutions.** A common wrong answer describes parallel lines, which look similar on a small sketch but are never identical.
P12	Very hard	In the system 3x + 5y = 21 and 2x − y = 1, what is the value of y? 	3	From the second equation, y = 2x − 1. Substituting: 3x + 5(2x − 1) = 21 gives 13x = 26, so x = 2. Then y = 2(2) − 1 = **3**. A common wrong answer is x — check which variable the question asked for.

The one habit to build. Finish the problem. A system's solution is a point, and the question may want x, y, their sum, or the cost at the intersection — not the first number you find.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.

SAT Math 1.5 — Linear Inequalities

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Solving an inequality

solve exactly as for an equation — but flip the sign when you multiply or divide by a negative

Every move is the same as for an equation except one. Multiplying or dividing both sides by a negative number reverses the direction, and forgetting that is the classic error.



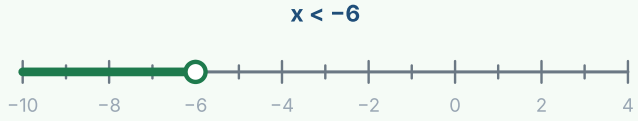
The flip is the whole difference from an equation. The open circle at −6 shows that the boundary itself is excluded by the strict inequality.

Example 1. **Solve 5 − 2x > 17.**

Solution.

−2x > 12, and dividing by −2 flips the sign.

x < **−6**.

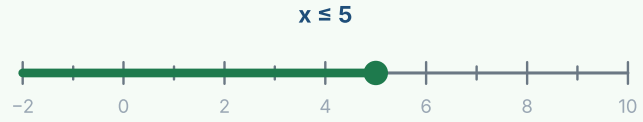


Example 2. **Solve 3x + 4 ≤ 19.**

Solution.

3x ≤ 15.

x ≤ **5**. No flip — we divided by a positive.

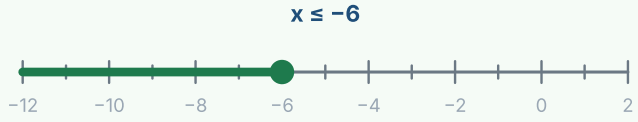


Example 3. **Solve −x/3 ≥ 2.**

Solution.

Multiply by −3 and flip.

x ≤ **−6**.



2 Translating the words

‘at least’ → ≥ | ‘at most’ / ‘no more than’ → ≤ | ‘more than’ → >

The wording decides whether the boundary counts. ‘Greater than $50’ excludes $50 exactly, so the least whole number of sales is one past the boundary.

	does the boundary value itself count?
at least 20	m ≥ 20 — 20 counts
at most 600 kg	L ≤ 600 — 600 counts
more than $50	p > 50 — 50 does not count

Two of these include their boundary and one does not. That single word is what separates 50 from 51 in the answer.

Example 1. **A vendor spends $75 and sells at $2.50 a cup. How many cups for profit over $50?**

Solution.

2.5n − 75 > 50 gives n > 50.

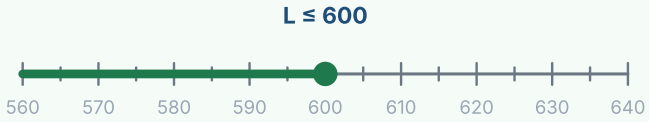
The least whole number is **51**, not 50.



Example 2. **A lift holds at most 600 kg. Write this as an inequality for the load L.**

Solution.

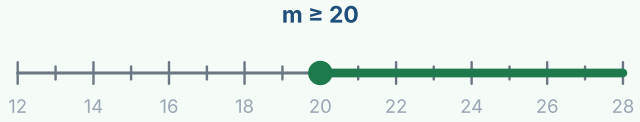
L ≤ **600** — 600 itself is allowed.



Example 3. **A club needs at least 20 members. Write it for membership m.**

Solution.

m ≥ **20**.



3 Interpreting a constant in context

ask what the units of each side are

In 3a + 7b ≤ 84, the left side is a cost in talens, so 84 is the **most** that can be spent — not a count of items. Checking units settles these questions immediately.

follow the units on each side

3a + 7b	3 and 7 are prices, so this is money
≤ 84	so 84 is a maximum amount of money
not a count	the units do not match items

Because the coefficients are prices, everything on that side is money. The number on the right has to be money too.

Example 1. **A stallholder buys a bags at 3 each and b jars at 7 each, with 3a + 7b ≤ 84. What is 84?**

Solution.
Both sides are costs, so 84 is the **maximum amount that can be spent**.

3a + 7b	a cost in talens
84	the most that can be spent

Example 2. **In 5x + 2y ≥ 40 where x is hours worked, what does 40 represent?**

Solution.
The **minimum** total the quantities must reach.

≥ 40	a floor, not a ceiling
the total must reach it	at least 40

Example 3. **Why is ‘the total number of items’ a wrong reading of 84?**

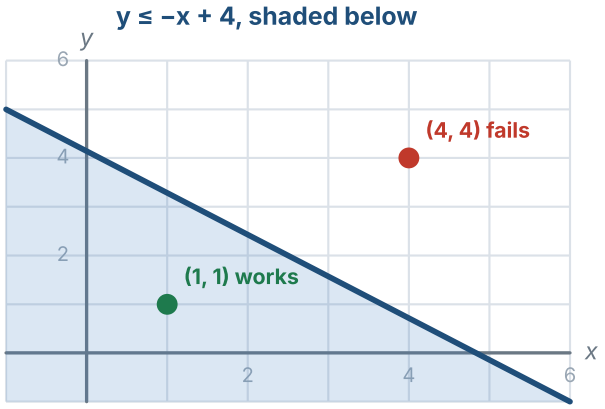
Solution.
The coefficients 3 and 7 are prices, so the product is money, not a count. **The units do not match.**

3 × bags	talens, not bags
a count would need	coefficients of 1

4 Graphing an inequality

solid line for ≤ or ≥ | dashed for < or > | test a point to decide which side

Draw the boundary line first, then pick any point not on it — (0, 0) is easiest — and see whether it satisfies the inequality. If it does, shade its side.



The line itself is included, so it is solid. (1, 1) satisfies the inequality and sits in the shaded region; (4, 4) fails and sits outside it.

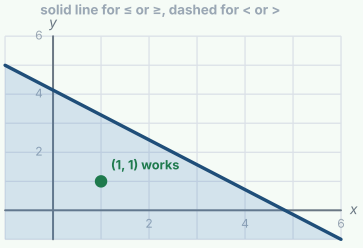
Example 1. **Describe the graph of y ≤ −x + 4.**

Solution.
A **solid** line through (0, 4) and (4, 0), with the region **below** shaded.



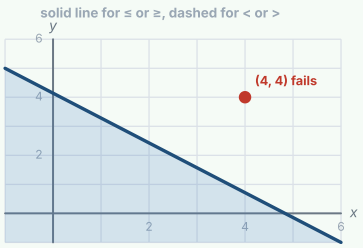
Example 2. **Does (1, 1) satisfy y ≤ −x + 4?**

Solution.
1 ≤ 3.
Yes, so it is in the shaded region.



Example 3. **Does (4, 4) satisfy it?**

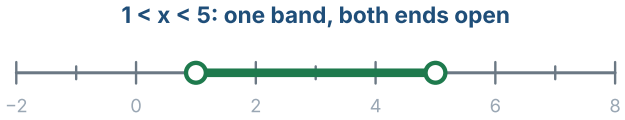
Solution.
4 ≤ 0 is false.
No — it lies above the line.



5 Compound inequalities

‘and’ → the overlap | ‘or’ → either region

Solve a three-part inequality by doing the same thing to all three parts at once. Keep the variable in the middle and remember that flipping applies to the whole chain.



An ‘and’ solution is a single stretch between two boundaries. Only the integers inside it — 2, 3 and 4 — qualify.

Example 1. **Solve $3 < 2x + 1 < 11$.**

Solution.

Subtract 1 throughout: $2 < 2x < 10$.

$1 < x < 5$.

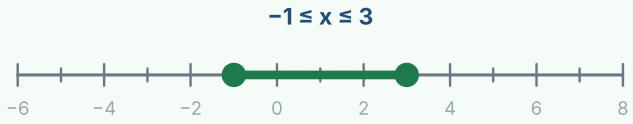


Example 2. **Solve $-4 \le 5 - 3x \le 8$.**

Solution.

Subtract 5: $-9 \le -3x \le 3$. Divide by -3 and flip both signs.

$-1 \le x \le 3$.



Example 3. **How many integers satisfy $1 < x < 5$?**

Solution.

$x = 2, 3, 4$.

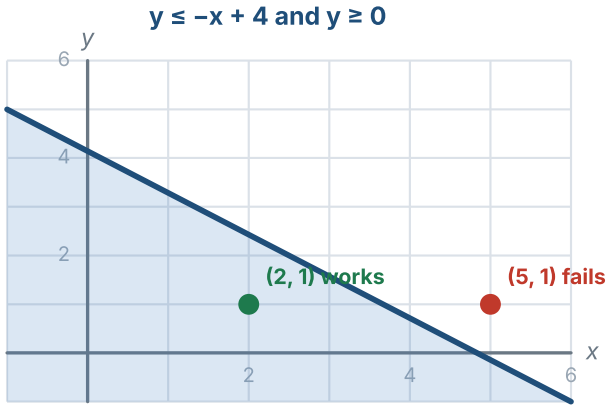
Three.



6 Systems of inequalities

the solution is the region satisfying every inequality at once

Shade each inequality and look for the overlap. To test whether a point is a solution, check it in every inequality — it only takes one failure to rule a point out.



(2, 1) satisfies both conditions; (5, 1) is above the line, so one failure is enough to exclude it.

Example 1. **Does (2, 1) satisfy y ≤ −x + 4 and y ≥ 0?**

Solution.

1 ≤ 2 ✓ and 1 ≥ 0 ✓.

Yes, both hold.

a solution must pass every inequality

y ≤ −x + 4	1 ≤ 2 ✓	✓ keep
y ≥ 0	1 ≥ 0 ✓	✓ keep

Example 2. **Does (5, 1) satisfy that system?**

Solution.

1 ≤ −1 is false.

No — one failure is enough.

one failure rules the point out

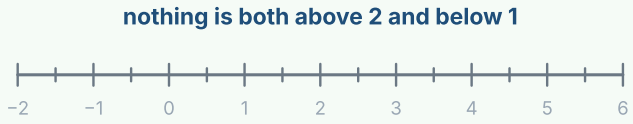
y ≤ −x + 4	1 ≤ −1 ×	× reject
y ≥ 0	1 ≥ 0 ✓	✓ keep

Example 3. **A system is y > 2 and y < 1. What is the solution region?**

Solution.

No value of y is both above 2 and below 1.

Empty — there is no solution.



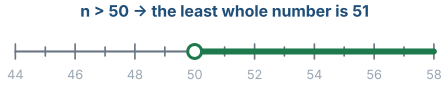
Part 1 — With Desmos (10 questions · every one is faster on the graph)

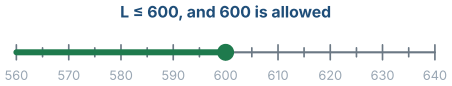
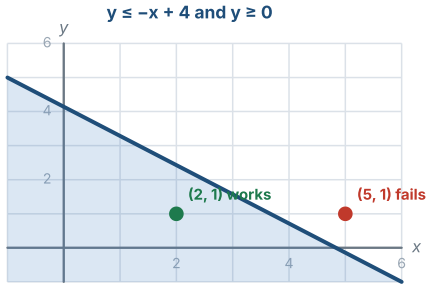
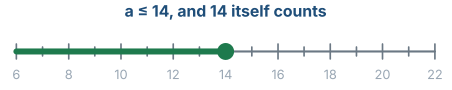
The SAT gives you Desmos on every maths question. Here it is **a region shader**. Type the inequality and Desmos shades the solution, boundary style included.

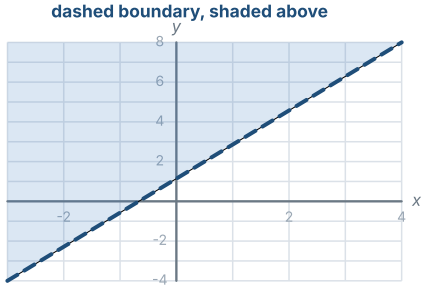
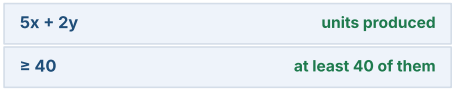
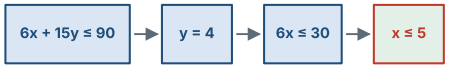
#	Question	Type into Desmos	Then	Answer
D1	Solve 5 − 2x > 17. What is the boundary value of x?	`5-2x=17`	Solve the equation, then decide the direction.	-6
D2	A vendor spends $75 and sells at $2.50 a cup. What is the break-even-plus-$50 cup count?	`(50+75)/2.5`	The boundary; the answer is one more.	50
D3	Solve 3x + 4 ≤ 19. What is the largest x?	`(19-4)/3`	Dividing by a positive keeps the sign.	5
D4	Solve 3 < 2x + 1 < 11. What are the two boundary values of x?	`2x+1=3`	Do each end separately.	1 and 5
D5	Solve −4 ≤ 5 − 3x. What is the boundary value of x?	`(5+4)/3`	Flip the sign when dividing by −3.	3
D6	Does (1, 1) satisfy y ≤ −x + 4?	`-1+4`	Compare the result with 1.	yes
D7	Does (4, 4) satisfy y ≤ −x + 4?	`-4+4`	4 is not less than or equal to 0.	no
D8	How many integers satisfy 1 < x < 5?	`4-1-0`	Count 2, 3 and 4.	3
D9	Graph y ≤ −x + 4 and shade. Which side contains the origin?	`y<=-x+4`	Desmos shades automatically.	below
D10	If 3a + 7b ≤ 84 and b = 6, what is the greatest whole value of a?	`(84-42)/3`	Substitute, then solve.	14

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Decide whether the boundary value itself counts: 'more than 50' excludes 50, so the least whole number is 51.

#	Level	Question	Answer	Solution
P1	Hard	Which describes all values of x that satisfy 5 − 2x > 17? 	x < −6	−2x > 12, and dividing by −2 **reverses** the inequality. x < **−6**. A common wrong answer is what you get by forgetting to flip — the single most common error in this topic.
P2	Hard	A lemonade vendor spends $75 on supplies each morning and sells lemonade for $2.50 per cup. What is the least whole number of cups the vendor must sell so that the day's profit is greater than $50? 	51	2.5n − 75 > 50 gives 2.5n > 125, so n > 50. The least whole number strictly greater than 50 is **51**. A common wrong answer is the boundary itself, which 'greater than' excludes.

#	Level	Question	Answer	Solution
P3	Hard	A stallholder buys a bags of rice at 3 talens each and b jars of honey at 7 talens each, where 3a + 7b ≤ 84. What is the best interpretation of 84? 3a + 7b — prices × quantities = money ≤ 84 — a maximum amount of money not a count of items — the units would not match	**The maximum number of talens the stallholder can spend.**	Both sides are amounts of money, and ≤ caps the left side. **The most that can be spent.** The trap five of your class took: 84 cannot be a count, because 3 and 7 are prices.
P4	Hard	Which inequality represents 'a lift holds at most 600 kilograms', where L is the load? 	**L ≤ 600**	'At most' means it may reach 600 but not exceed it. **L ≤ 600.** A common wrong answer would forbid a load of exactly 600 kg.
P5	Hard	If 3 < 2x + 1 < 11, how many integer values can x take? 	**3**	Subtracting 1 throughout gives 2 < 2x < 10, so 1 < x < 5. The integers are 2, 3 and 4, so there are **3** of them. The endpoints are excluded.
P6	Hard	Which point satisfies the system y ≤ −x + 4 and y ≥ 0? 	**(2, 1)**	Test each: at (2, 1), 1 ≤ 2 ✓ and 1 ≥ 0 ✓. **(2, 1) works.** (5, 1) fails the first; (1, −2) fails the second.
P7	Very hard	If −4 ≤ 5 − 3x ≤ 8, which describes all possible values of x? 	**−1 ≤ x ≤ 3**	Subtract 5 throughout: −9 ≤ −3x ≤ 3. Dividing by −3 flips both signs and swaps the ends. **−1 ≤ x ≤ 3.** A common wrong answer is what you get without flipping.
P8	Very hard	In the inequality 3a + 7b ≤ 84, b = 6. What is the greatest possible whole-number value of a? 	**14**	3a + 42 ≤ 84 gives 3a ≤ 42. a ≤ 14, so the greatest whole value is **14** — the boundary is allowed because the sign is ≤.

#	Level	Question	Answer	Solution
P9	Very hard	A student must score at least 80 as an average over four tests. After three tests the scores are 72, 85 and 79. What is the minimum score needed on the fourth test? 	84	The total must be at least 4 × 80 = 320, and the first three sum to 236. 320 − 236 = **84**.
P10	Very hard	The graph of a linear inequality shows a **dashed** boundary line with the region above it shaded. Which inequality could it represent? 	y > 2x + 1	Dashed means the boundary is **not** included, so the sign is strict; shading above means y is greater. **y > 2x + 1.** A common wrong answer would need a solid line.
P11	Very hard	A company needs at least 40 units produced, using machine A at 5 units per hour and machine B at 2 units per hour. If x and y are the hours each runs, which inequality models this? 	5x + 2y ≥ 40	'At least 40 units' means the total production must reach 40 or more. **5x + 2y ≥ 40.** A common wrong answer counts hours rather than units.
P12	Very hard	A caterer has at most 90 talens to spend on x trays of bread at 6 talens each and y trays of cheese at 15 talens each. If 4 trays of cheese are bought, what is the greatest whole number of bread trays the caterer can buy? 	5	6x + 15y ≤ 90 with y = 4 gives 6x + 60 ≤ 90, so 6x ≤ 30. x ≤ 5, and the greatest whole number is **5**.

The one habit to build. Flip the inequality sign whenever you multiply or divide by a negative. It is the only rule that differs from solving an equation, and it is where the marks go.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.

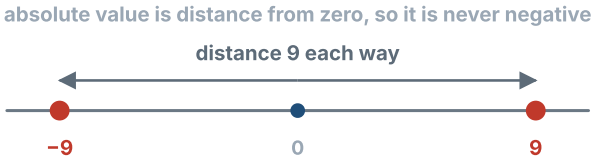
SAT Math 1.6 — Absolute Value

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 What absolute value means

|a| is the distance from zero, so it is never negative

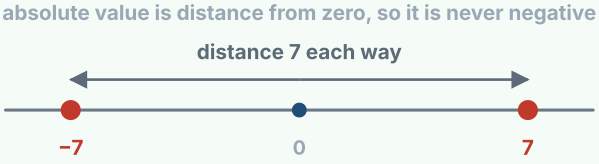
Distance has no direction, which is why |−7| = 7. An equation |x| = c with c negative has no solution at all, because no distance is negative.



Both −9 and 9 sit nine units from zero, so both satisfy |x| = 9. Nothing sits a negative distance away, which is why |x| = −4 has no solution.

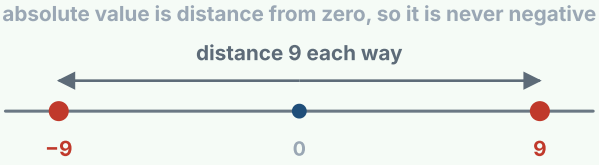
Example 1. **Evaluate |−7| + |3|.**

Solution.
7 + 3 = **10**.



Example 2. **Solve |x| = 9.**

Solution.
x is 9 away from zero in either direction.
x = **9 or −9**.



Example 3. **Solve |x| = −4.**

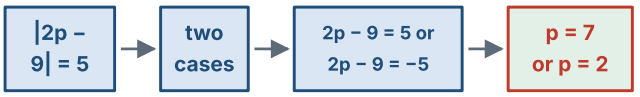
Solution.
No solution — an absolute value can never be negative.

|x| = −4	**a distance cannot be negative**
nothing to split	**no solution**

2 Solving |ax + b| = c

split into two equations: ax + b = c and ax + b = −c

The bars hide a plus-or-minus. Write both cases before solving either, and check c is not negative first.

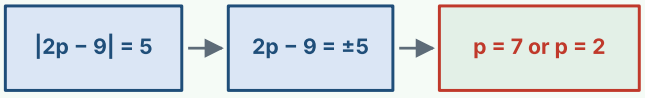


both cases are real solutions — neither is a spare

One set of bars, two equations. Taking only the positive case is what loses half the answer.

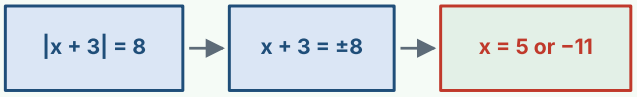
Example 1. **Solve |2p − 9| = 5.**

Solution.
2p − 9 = 5 gives p = 7; 2p − 9 = −5 gives p = 2.
p = 7 or 2, and their sum is 9.



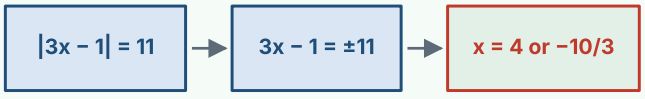
Example 2. **Solve |x + 3| = 8.**

Solution.
x + 3 = 8 or x + 3 = −8.
x = **5 or −11**.



Example 3. **Solve |3x − 1| = 11.**

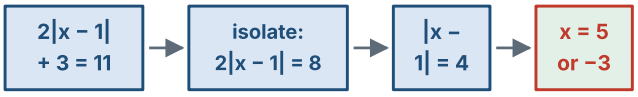
Solution.
3x − 1 = 11 gives x = 4; 3x − 1 = −11 gives x = −10/3.
x = **4 or −10/3**.



3 Isolate the bars first

get |...| alone on one side before splitting

If something is added to or multiplying the absolute value, undo that first. Splitting too early produces two wrong equations instead of one right pair.

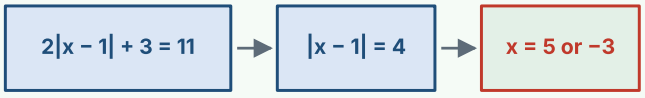


the +3 and the 2 apply to the whole distance

The 2 and the +3 sit outside the bars, so they act on the whole distance. Only once the bars stand alone can the split be made.

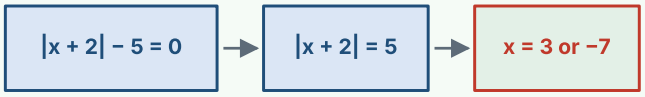
Example 1. **Solve $2|x - 1| + 3 = 11$.**

Solution.
Isolate: $2|x - 1| = 8$, so $|x - 1| = 4$.
$x - 1 = \pm 4$, giving $x =$ **5 or −3**.



Example 2. **Solve $|x + 2| - 5 = 0$.**

Solution.
$|x + 2| = 5$.
$x =$ **3 or −7**.



Example 3. **Why not split $2|x - 1| + 3 = 11$ straight away?**

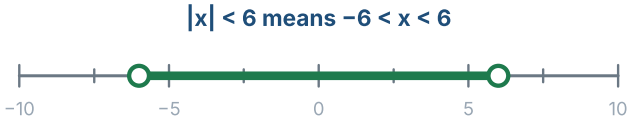
Solution.
The +3 and the 2 are outside the bars, so they apply to the whole distance. Splitting first would attach them to only one side and **both answers would be wrong**.

$2\|x - 1\| + 3 = 11$	the 2 and 3 are outside
isolate first	$\|x - 1\| = 4$
then split	$x - 1 = \pm 4$

4 Absolute value inequalities: the ‘and’ case

$|x| < c$ means $-c < x < c$ — one interval, between

Less than means close to zero, so the solution is a single band around the centre. Write it as a compound inequality and solve all three parts at once.



Everything within six units of zero, in either direction: one unbroken band, with both ends open because the inequality is strict.

Example 1. **Solve $|x| < 6$.**

Solution.

$-6 < x < 6$.

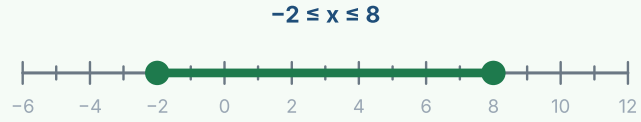


Example 2. **Solve $|x - 3| \le 5$.**

Solution.

$-5 \le x - 3 \le 5$, so add 3 throughout.

$-2 \le x \le 8$.



Example 3. **Solve $|2x + 1| < 7$.**

Solution.

$-7 < 2x + 1 < 7$ gives $-8 < 2x < 6$.

$-4 < x < 3$.



5 Absolute value inequalities: the ‘or’ case

$|x| > c$ means $x < -c$ or $x > c$ — two separate pieces

Greater than means far from zero, so the solution splits into two rays with a gap in the middle. It can never be written as a single interval.

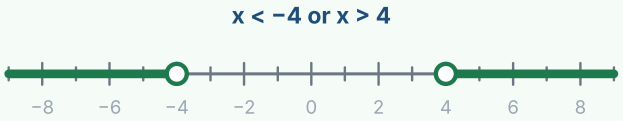


Two pieces with a hole between them. Writing this as one interval is impossible — the middle values fail the inequality.

Example 1. **Solve $|x| > 4$.**

Solution.

$x < -4$ or $x > 4$.

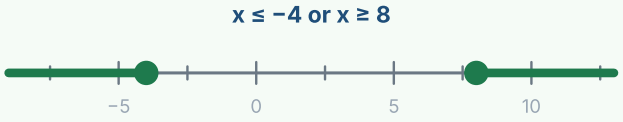


Example 2. **Solve $|x - 2| \geq 6$.**

Solution.

$x - 2 \geq 6$ or $x - 2 \leq -6$.

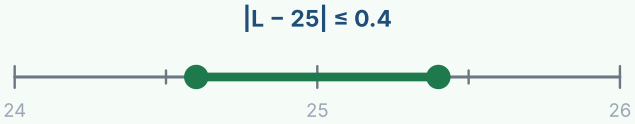
$x \geq 8$ or $x \leq -4$.



Example 3. **A machine part must be within 0.4 mm of 25 mm. Write that as an inequality.**

Solution.

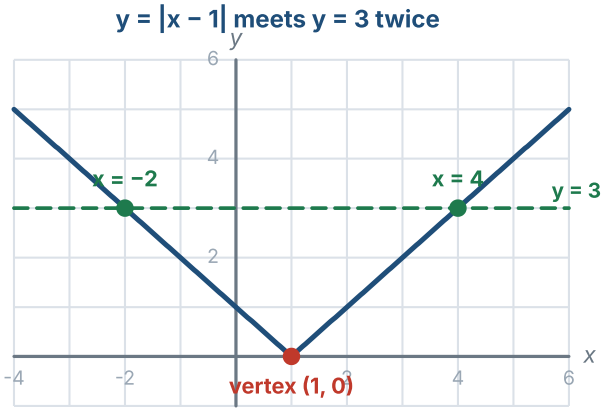
$|L - 25| \leq 0.4$, which means $24.6 \leq L \leq 25.4$.



6 The V-shaped graph

$y = |x - h| + k$ is a V with vertex (h, k)

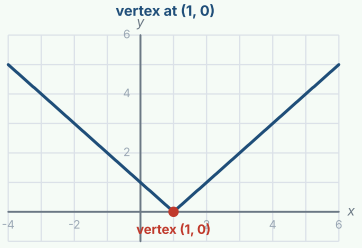
The graph folds upward at the vertex. A horizontal line cuts it twice above the vertex, once at the vertex, and not at all below — which is how the number of solutions is decided.



The line $y = 3$ sits above the vertex, so it crosses both arms, at $x = -2$ and $x = 4$. Lowering it to the vertex would give one crossing; below, none.

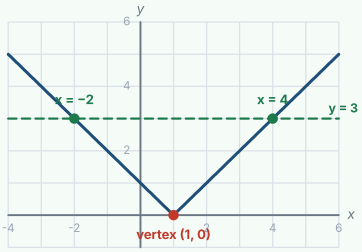
Example 1. **Find the vertex of $y = |x - 1|$.**

Solution.
(1, 0).



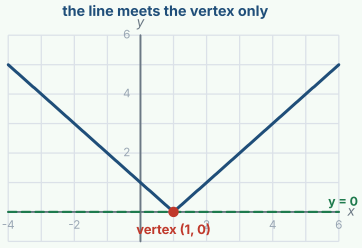
Example 2. **How many solutions does $|x - 1| = 3$ have?**

Solution.
The line $y = 3$ is above the vertex, so it cuts the V twice.
Two: $x = 4$ and $x = -2$.



Example 3. **How many solutions does $|x - 1| = 0$ have?**

Solution.
The line passes exactly through the vertex.
One: $x = 1$.



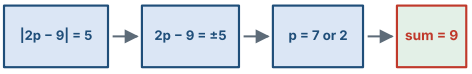
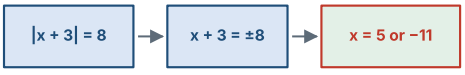
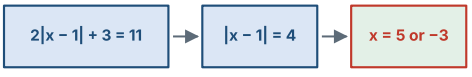
Part 1 — With Desmos (10 questions · every one is faster on the graph)

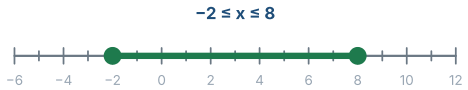
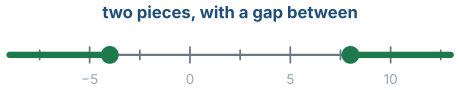
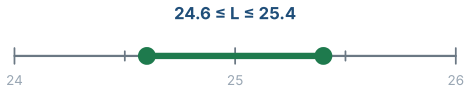
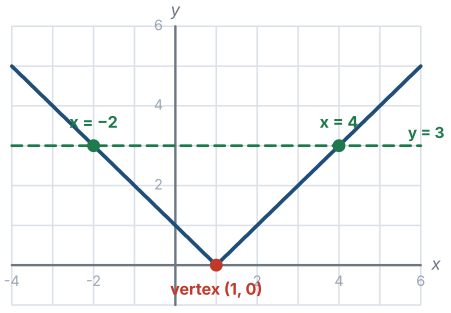
The SAT gives you Desmos on every maths question. Here it is **a V-grapher**. Type abs(x − 1) alongside a horizontal line and count the crossings to see how many solutions there are.

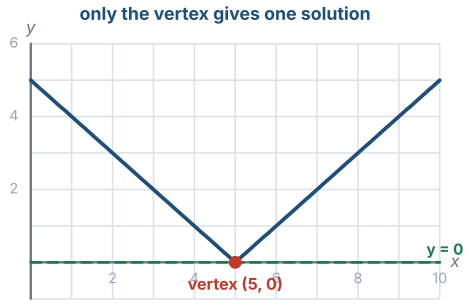
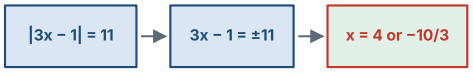
#	Question	Type into Desmos	Then	Answer
D1	What is $\|-7\| + \|3\|$?	`abs(-7)+abs(3)`	Use abs() in Desmos.	10
D2	Solve $\|2p - 9\| = 5$. What is the sum of the solutions?	`7+2`	Graph both sides and read the two crossings.	9
D3	Solve $\|x + 3\| = 8$. What is the smaller solution?	`-8-3`	Two cases, two answers.	-11
D4	Solve $2\|x - 1\| + 3 = 11$. What is the larger solution?	`4+1`	Isolate the bars before splitting.	5
D5	Solve $\|x - 3\| \le 5$. What is the upper bound of x?	`5+3`	The 'and' case.	8
D6	Solve $\|2x + 1\| < 7$. What is the lower bound of x?	`(-7-1)/2`	Three parts at once.	-4
D7	Solve $\|x - 2\| \ge 6$. What is the smaller boundary?	`-6+2`	The 'or' case.	-4
D8	A part must be within 0.4 mm of 25 mm. What is the smallest allowed length?	`25-0.4`	Centre minus the tolerance.	24.6
D9	Graph $y = \|x - 1\|$. What is the x-coordinate of the vertex?	`y=abs(x-1)`	The V folds at the vertex.	1
D10	How many solutions does $\|x - 1\| = 3$ have?	`y=abs(x-1) and y=3`	Count the intersections.	2

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Decide first whether the inequality is the 'and' case (one band) or the 'or' case (two pieces).

#	Level	Question	Answer	Solution
P1	Hard	The price p, in talens, of a lantern satisfies $\|2p - 9\| = 5$. What is the sum of all possible values of p? 	9	$2p - 9 = 5$ gives $p = 7$, and $2p - 9 = -5$ gives $p = 2$. $7 + 2 = \mathbf{9}$. Taking only the positive case and answering 7 is the usual error.
P2	Hard	Solve $\|x + 3\| = 8$. Give both solutions. 	x = 5 and x = −11	$x + 3 = 8$ gives $x = 5$; $x + 3 = -8$ gives $x = -11$. Both are valid: the bars hide a plus-or-minus.
P3	Hard	Solve $2\|x - 1\| + 3 = 11$. Give both solutions. 	x = 5 and x = −3	Isolate first: $2\|x - 1\| = 8$, so $\|x - 1\| = 4$. $x - 1 = \pm 4$ gives **5 or −3**. Splitting before isolating gives the wrong pair.

#	Level	Question	Answer	Solution
P4	Hard	Solve the inequality $\|x - 3\| \le 5$. Write your answer as a compound inequality. 	$-2 \le x \le 8$	Less than means a single band: $-5 \le x - 3 \le 5$. Adding 3 throughout gives **$-2 \le x \le 8$**.
P5	Hard	Solve the inequality $\|x - 2\| \ge 6$. Describe all values of x. 	$x \le -4$ or $x \ge 8$	Greater than means two separate pieces: $x - 2 \ge 6$ or $x - 2 \le -6$. **$x \ge 8$ or $x \le -4$.** This cannot be written as one interval.
P6	Hard	A machine part must be within 0.4 millimetres of a target length of 25 millimetres. Write an absolute-value inequality for the acceptable length L. 	$\|L - 25\| \le 0.4$	‘Within 0.4 of 25’ is a distance statement. **$\|L - 25\| \le 0.4$**, which unpacks to $24.6 \le L \le 25.4$.
P7	Very hard	How many solutions does the equation $\|3x - 6\| = -2$ have? Explain briefly. $\|3x - 6\| = -2$ — a distance cannot be negative check c first — no solutions, no work needed	none	An absolute value is a distance and can never be negative, so no value of x can satisfy it. **No solutions.** There is nothing to split — checking the right-hand side first saves the whole calculation.
P8	Very hard	Solve $\|2x + 1\| < 7$. Write your answer as a compound inequality. 	$-4 < x < 3$	$-7 < 2x + 1 < 7$ gives $-8 < 2x < 6$. Dividing by 2 gives **$-4 < x < 3$**.
P9	Very hard	The graph of $y = \|x - 1\|$ is drawn together with the horizontal line $y = 3$. At how many points do they intersect, and at which x-values? 	two, at $x = 4$ and $x = -2$	y = 3 sits above the vertex (1, 0), so it cuts both arms of the V. $\|x - 1\| = 3$ gives $x - 1 = \pm 3$, so x = **4 or −2**.

#	Level	Question	Answer	Solution
P10	Very hard	For what value of c does the equation $\|x - 5\| = c$ have exactly one solution? 	0	A horizontal line meets the V twice above the vertex and not at all below it. Exactly one intersection happens only at the vertex itself, so c = **0**.
P11	Very hard	Solve $\|3x - 1\| = 11$. Give both solutions. 	x = 4 and x = −10/3	$3x - 1 = 11$ gives $3x = 12$, so $x = 4$. $3x - 1 = -11$ gives $3x = -10$, so $x = -10/3$.
P12	Very hard	A thermostat keeps a room within 1.5°C of 21°C. Write an absolute-value inequality for the temperature T, and state the range of acceptable temperatures. 	$\|T - 21\| \le 1.5$, so $19.5 \le T \le 22.5$	'Within 1.5 of 21' gives $\|T - 21\| \le 1.5$. Unpacking: $-1.5 \le T - 21 \le 1.5$, so $19.5 \le T \le 22.5$.

The one habit to build. Isolate the bars before splitting into two cases, and check the right-hand side is not negative — if it is, there is no solution and no work to do.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.